NEXT ENGINEER · _SHEERAT_

COMPLETE TEXTBOOK

Calculus and Linear Algebra

AAS103D

Full syllabus, unit by unit: theory in question-answer form, proofs, step-by-step solved problems and 2-mark questions

UNIT 1 Differential Calculus of Several Variables

UNIT 2 Multiple Integrals

UNIT 3 Vector Calculus

UNIT 4 Matrices

UNIT 5 Introduction to Vector Spaces

Program: B.Tech **Branches:** CSE, IT, CSE-AIML, CSE-DS, AIDS **Semester:** 1 **Credits:** 4

L-T-P: 3-1-0 **Scheme:** AKTU B.Tech 1st year, CSE stream, 2026-27 onwards

First edition, 2026

Contents

How to Use This Book

Unit 1 Differential Calculus of Several Variables

Part 1: Limits, Continuity and Partial Derivatives
Part 2: Euler's Theorem for Homogeneous Functions
Part 3: Composite Functions, Total Derivative and Change of Variables
Part 4: Jacobians
Part 5: Maxima and Minima
Part 6: Lagrange's Method of Undetermined Multipliers
Part 7: Errors and Approximations

Unit 2 Multiple Integrals

Part 1: Double Integrals
Part 2: Change of Order of Integration
Part 3: Change of Variables and Polar Coordinates
Part 4: Area by Double Integration
Part 5: Triple Integrals and Volume
Part 6: Beta and Gamma Functions
Part 7: Dirichlet's Integral

Unit 3 Vector Calculus

Part 1: Gradient and Directional Derivative
Part 2: Divergence and Curl
Part 3: Line Integrals and Work Done
Part 4: Surface and Volume Integrals; Green's Theorem
Part 5: Stokes' Theorem
Part 6: Gauss's Divergence Theorem

Unit 4 Matrices

Part 1: Special Types of Matrices
Part 2: Elementary Transformations, Inverse and Rank
Part 3: Systems of Linear Equations
Part 4: Eigenvalues and Eigenvectors
Part 5: Similarity and Diagonalization

Unit 5 Introduction to Vector Spaces

Part 1: Fields and Vector Spaces
Part 2: Subspaces
Part 3: Linear Combinations and Span
Part 4: Linear Dependence and Independence
Part 5: Basis and Dimension
Part 6: Sum and Direct Sum of Subspaces

Model Question Paper with Answer Guide

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Last-Day Revision Checklist

NEXT ENGINEER
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How to Use This Book

This book covers the complete syllabus of Engineering Mathematics I (Calculus and Linear Algebra), unit by unit, in the question-and-answer style of the university examination. Every method is explained from the beginning and followed by fully solved examples, so you can study from this book alone. It is written in original words for Next Engineer.

How each unit is organised

- A short **overview** tells you what the unit is about and which question types are asked most.
- The unit is divided into **Parts**. Each Part covers a few syllabus topics.
- Inside each Part, theory and problems are written as **long-answer questions** (*Que 1.1*, *Que 1.2*, ...). Theory answers give the definition, the result and its proof or reasoning. Problem answers show every step, the way you should write it in the exam.
- Each unit ends with **2-mark short-answer questions** with answers and a set of **practice problems** with answers.

The exam pattern (theory, 70 marks, 3 hours)

Section	What is asked	Marks
A	7 short questions, all compulsory, 2 marks each	14
B	Attempt any 3 out of 5 questions, 7 marks each	21
C	One question per unit, each with two parts; attempt one part of each, 7 marks each	35

Internal assessment carries another 30 marks.

How to score in mathematics

1. Write the **formula or theorem first**, then substitute. Marks are given for the method even if the arithmetic slips.
2. Never skip steps in a 7-mark answer. Examiners mark step by step.
3. For multiple integrals, change of order, Green's, Stokes' and Gauss's theorems, **draw the region**.
4. **Box the final answer** and write units where they apply.
5. Check your answer: substitute back into equations, check that a matrix times its inverse gives I, check that the sum of eigenvalues equals the trace.

Prescribed books (for further reading)

- Erwin Kreyszig, *Advanced Engineering Mathematics*, John Wiley & Sons.
- B. V. Ramana, *Higher Engineering Mathematics*, Tata McGraw Hill.
- B. S. Grewal, *Higher Engineering Mathematics*, Khanna Publishers.
- R. K. Jain and S. R. K. Iyengar, *Advanced Engineering Mathematics*, Narosa Publishing House.
- Peter V. O'Neil, *Advanced Engineering Mathematics*, Thomson Learning.
- G. B. Thomas and R. L. Finney, *Calculus and Analytic Geometry*, Pearson.
- D. Poole, *Linear Algebra: A Modern Introduction*, Brooks/Cole.

- T. Veerarajan, *Engineering Mathematics for First Year*, Tata McGraw Hill.

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UNIT 1

Differential Calculus of Several Variables

Unit overview

In school calculus a quantity depends on one variable. In engineering it usually depends on several: the temperature of a plate depends on x and y, the volume of a gas on pressure and temperature, the cost of a design on many dimensions. This unit extends differentiation to such functions.

Topics in this unit: limits and continuity of functions of two variables, partial derivatives, composite functions and the chain rule, total derivative, Euler's theorem for homogeneous functions, change of variables, Jacobians, maxima and minima of functions of two variables, Lagrange's method of undetermined multipliers, errors and approximations.

Most asked: Euler's theorem problems (first and second order), total derivative and chain rule proofs, Jacobians (including functional dependence and $JJ' = 1$), maxima and minima using $rt - s^2$, Lagrange multipliers, and percentage error.

Part 1: Limits, Continuity and Partial Derivatives

Que 1.1 **Define the limit and continuity of a function of two variables. Show that $\lim_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$ does not exist, and that $f(x,y) = \dfrac{x^2y}{x^2+y^2}$, $f(0,0) = 0$, is continuous at the origin.**

ANSWER

Limit. A function $f(x,y)$ tends to the limit L as $(x,y) \to (a,b)$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x,y) - L| < \varepsilon \quad \text{whenever} \quad 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta.$$

In one variable, x can approach a only from the left or the right. In two variables, (x,y) can approach (a,b) along **infinitely many paths** (straight lines, parabolas, and so on). The limit exists only if f approaches the **same value along every path**. So, to prove that a limit does **not** exist, it is enough to find two paths that give different values.

Continuity. f is continuous at (a,b) if

1. $f(a,b)$ is defined,
2. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists, and
3. the limit equals $f(a,b)$.

Example 1. Approach $(0,0)$ along the line $y = mx$:

$$\frac{xy}{x^2+y^2} = \frac{x \cdot mx}{x^2+m^2x^2} = \frac{m}{1+m^2}.$$

This depends on m: along $y = x$ the value is $\frac{1}{2}$, along $y = 0$ it is 0. Different paths give different values, so **the limit does not exist**.

Example 2. For $(x, y) \neq (0, 0)$, since $x^2 \leq x^2 + y^2$,

$$|f(x,y) - 0| = \frac{x^2|y|}{x^2+y^2} \leq |y| \leq \sqrt{x^2+y^2}.$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $\sqrt{x^2+y^2} < \delta$ gives $|f(x,y)| < \varepsilon$. So $\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$, and f **is continuous at the origin**.

Note. Checking straight lines only is not enough to prove that a limit exists. For $\frac{x^2y}{x^4+y^2}$ every line $y = mx$ gives the limit 0, but along the parabola $y = mx^2$ the value is $\frac{m}{1+m^2}$, so the limit does not exist.

Que 1.2 **Define partial derivatives. Explain their geometric meaning and the notation for higher-order partial derivatives.**

ANSWER

Let $z = f(x, y)$. The **partial derivative of f with respect to x** is the ordinary derivative of f with respect to x, treating y as a constant:

$$\frac{\partial f}{\partial x} = f_x = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}.$$

Similarly,

$$\frac{\partial f}{\partial y} = f_y = \lim_{k\to 0} \frac{f(x,y+k) - f(x,y)}{k}.$$

Geometric meaning. $z = f(x, y)$ is a surface. The plane $y = y_0$ cuts it in a curve; $f_x(x_0, y_0)$ is the slope of the tangent to this curve at (x_0, y_0), that is, the rate of change of z in the x-direction. f_y is the slope in the y-direction.

Higher-order partial derivatives

$$\frac{\partial^2 f}{\partial x^2} = f_{xx}, \qquad \frac{\partial^2 f}{\partial y^2} = f_{yy}, \qquad \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right), \qquad \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

Schwarz's (Clairaut's) theorem: if f_{xy} and f_{yx} are continuous, then $f_{xy} = f_{yx}$. For all the functions met in this course the mixed partial derivatives are equal.

In the theory of maxima and minima the following notation is standard:

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}, \quad r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2}.$$

Que 1.3 If $u = \log(x^2 + y^2)$, verify that $u_{xy} = u_{yx}$ and show that $u_{xx} + u_{yy} = 0$.

ANSWER

$$u_x = \frac{2x}{x^2 + y^2}, \qquad u_y = \frac{2y}{x^2 + y^2}.$$

Mixed derivatives.

$$u_{yx} = \frac{\partial}{\partial y}\left(\frac{2x}{x^2 + y^2}\right) = -\frac{2x \cdot 2y}{(x^2 + y^2)^2} = -\frac{4xy}{(x^2 + y^2)^2},$$

$$u_{xy} = \frac{\partial}{\partial x}\left(\frac{2y}{x^2 + y^2}\right) = -\frac{4xy}{(x^2 + y^2)^2}.$$

Hence $u_{xy} = u_{yx}$.

Laplace's equation. By the quotient rule,

$$u_{xx} = \frac{2(x^2 + y^2) - 2x \cdot 2x}{(x^2 + y^2)^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}, \qquad u_{yy} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}.$$

Adding, $u_{xx} + u_{yy} = 0$. A function that satisfies this equation is called **harmonic**.

Que 1.4 If $u = (x^2 + y^2 + z^2)^{-1/2}$, show that $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$.

ANSWER

Let $r^2 = x^2 + y^2 + z^2$, so $u = r^{-1}$ and $\dfrac{\partial r}{\partial x} = \dfrac{x}{r}$.

$$\frac{\partial u}{\partial x} = -\frac{1}{r^2} \cdot \frac{x}{r} = -\frac{x}{r^3}.$$

$$\frac{\partial^2 u}{\partial x^2} = -\left[\frac{1}{r^3} + x \cdot (-3) r^{-4} \cdot \frac{x}{r}\right] = -\frac{1}{r^3} + \frac{3x^2}{r^5}.$$

By symmetry,

$$\frac{\partial^2 u}{\partial y^2} = -\frac{1}{r^3} + \frac{3y^2}{r^5}, \qquad \frac{\partial^2 u}{\partial z^2} = -\frac{1}{r^3} + \frac{3z^2}{r^5}.$$

Adding,

$$\nabla^2 u = -\frac{3}{r^3} + \frac{3(x^2 + y^2 + z^2)}{r^5} = -\frac{3}{r^3} + \frac{3r^2}{r^5} = 0.$$

NEXT ENGINEER sheerat_

Que 1.5 **If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that**

(i) $\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} + \dfrac{\partial u}{\partial z} = \dfrac{3}{x+y+z}$, and (ii) $\left(\dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}\right)^2 u = -\dfrac{9}{(x+y+z)^2}$.

ANSWER

(i) Let $D = x^3 + y^3 + z^3 - 3xyz$. Then

$$u_x = \frac{3x^2 - 3yz}{D}, \quad u_y = \frac{3y^2 - 3zx}{D}, \quad u_z = \frac{3z^2 - 3xy}{D}.$$

$$u_x + u_y + u_z = \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{x^3 + y^3 + z^3 - 3xyz}.$$

Using the identity $x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)$,

$$u_x + u_y + u_z = \frac{3}{x+y+z}.$$

(ii) Write $L = \dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}$. Then

$$L^2 u = L\left(\frac{3}{x+y+z}\right) = -\frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} = -\frac{9}{(x+y+z)^2}.$$

Part 2: Euler's Theorem for Homogeneous Functions

Que 1.6 **Define a homogeneous function. State and prove Euler's theorem on homogeneous functions.**

ANSWER

Homogeneous function. $f(x, y)$ is homogeneous of degree n if

$$f(tx, ty) = t^n f(x, y) \quad \text{for all } t > 0.$$

Equivalently, f can be written as $f(x, y) = x^n \, \phi\!\left(\dfrac{y}{x}\right)$.

Examples: $x^3 + 3x^2 y - y^3$ is homogeneous of degree 3; $\dfrac{x^2 + y^2}{x + y}$ has degree 1; $\dfrac{\sqrt{x} + \sqrt{y}}{x + y}$ has degree $-\frac{1}{2}$; $\sin^{-1}\dfrac{y}{x}$ has degree 0.

Euler's theorem. If $u = f(x, y)$ is a homogeneous function of degree n with continuous partial derivatives, then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u.$$

Proof. Write $u = x^n \phi(v)$ with $v = \dfrac{y}{x}$. Then $\dfrac{\partial v}{\partial x} = -\dfrac{y}{x^2}$ and $\dfrac{\partial v}{\partial y} = \dfrac{1}{x}$.

$$\frac{\partial u}{\partial x} = n x^{n-1}\phi(v) + x^n \phi'(v)\left(-\frac{y}{x^2}\right) \;\Rightarrow\; x\frac{\partial u}{\partial x} = n x^n \phi(v) - x^{n-1} y\,\phi'(v).$$

$$\frac{\partial u}{\partial y} = x^n \phi'(v)\cdot\frac{1}{x} \;\Rightarrow\; y\frac{\partial u}{\partial y} = x^{n-1} y\,\phi'(v).$$

Adding, $x u_x + y u_y = n x^n \phi(v) = n u$. ∎

The theorem extends to any number of variables: for $u(x, y, z)$ homogeneous of degree n, $x u_x + y u_y + z u_z = n u$.

Que 1.7 **Prove that if u is homogeneous of degree n, then $x^2 u_{xx} + 2xy\,u_{xy} + y^2 u_{yy} = n(n-1)u$. Also prove the two corollaries used when u itself is not homogeneous but $f(u)$ is.**

ANSWER

Second-order result. By Euler's theorem,

$$x u_x + y u_y = n u. \tag{1}$$

Differentiate (1) partially with respect to x:

$$u_x + x u_{xx} + y u_{yx} = n u_x \;\Rightarrow\; x u_{xx} + y u_{xy} = (n-1) u_x. \tag{2}$$

Differentiate (1) with respect to y:

$$x u_{xy} + u_y + y u_{yy} = n u_y \;\Rightarrow\; x u_{xy} + y u_{yy} = (n-1) u_y. \tag{3}$$

Multiply (2) by x, (3) by y and add:

$$x^2 u_{xx} + 2xy\,u_{xy} + y^2 u_{yy} = (n-1)(x u_x + y u_y) = n(n-1)u. \;\blacksquare$$

Corollary 1. If $z = f(u)$ is homogeneous of degree n in x and y (although u is not), then

$$x u_x + y u_y = n\frac{f(u)}{f'(u)} = g(u).$$

Proof: by Euler's theorem applied to z, $x z_x + y z_y = n z$, that is $x f'(u) u_x + y f'(u) u_y = n f(u)$. Divide by $f'(u)$.

Corollary 2. With $g(u) = n\dfrac{f(u)}{f'(u)}$ as above,

$$x^2 u_{xx} + 2xy\,u_{xy} + y^2 u_{yy} = g(u)\left[g'(u) - 1\right].$$

NEXT ENGINEER _sheerat_

Proof: differentiate $xu_x + yu_y = g(u)$ with respect to x and y, multiply by x and y, and add, exactly as above. The right side becomes $g'(u)(xu_x + yu_y) - (xu_x + yu_y) = g(u)[g'(u) - 1]$.

Que 1.8 **If $u = \sin^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, prove that**

(i) $xu_x + yu_y = \frac{1}{2}\tan u$, and (ii) $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = -\dfrac{\sin u \cos 2u}{4\cos^3 u}$.

ANSWER

u is not homogeneous, but

$$z = \sin u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

is homogeneous of degree $1 - \frac{1}{2} = \frac{1}{2}$ (numerator degree 1, denominator degree $\frac{1}{2}$).

(i) By Corollary 1 with $f(u) = \sin u$, $n = \frac{1}{2}$:

$$xu_x + yu_y = \frac{1}{2}\cdot\frac{\sin u}{\cos u} = \frac{1}{2}\tan u.$$

(ii) Here $g(u) = \frac{1}{2}\tan u$, so $g'(u) = \frac{1}{2}\sec^2 u$. By Corollary 2,

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = \frac{1}{2}\tan u\left(\frac{1}{2}\sec^2 u - 1\right) = \frac{\tan u}{4}\cdot\frac{1-2\cos^2 u}{\cos^2 u}.$$

Since $1 - 2\cos^2 u = -\cos 2u$,

$$= -\frac{\sin u \cos 2u}{4\cos^3 u}. \blacksquare$$

Que 1.9 **If $u = \tan^{-1}\left(\dfrac{x^3+y^3}{x-y}\right)$, show that $xu_x + yu_y = \sin 2u$ and $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = \sin 4u - \sin 2u$.**

ANSWER

$z = \tan u = \dfrac{x^3+y^3}{x-y}$ is homogeneous of degree $3 - 1 = 2$.

By Corollary 1 with $f(u) = \tan u$, $f'(u) = \sec^2 u$, $n = 2$:

$$xu_x + yu_y = 2\frac{\tan u}{\sec^2 u} = 2\sin u\cos u = \sin 2u.$$

Now $g(u) = \sin 2u$ and $g'(u) = 2\cos 2u$. By Corollary 2,

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = \sin 2u\,(2\cos 2u - 1) = 2\sin 2u\cos 2u - \sin 2u = \sin 4u - \sin 2u. \blacksquare$$

NEXT ENGINEER _sheerat_

Part 3: Composite Functions, Total Derivative and Change of Variables

Que 1.10 **Explain the total differential and the chain rule for composite functions. If $u = x^2 + y^2$ with $x = at^2$, $y = 2at$, find $\frac{du}{dt}$ by the chain rule and verify it directly.**

ANSWER

Total differential. If $u = f(x, y)$, a small change dx in x and dy in y produces the change

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy.$$

Chain rule (one parameter). If $u = f(x, y)$ and both $x = \phi(t)$, $y = \psi(t)$, then u is a function of t alone (a **composite function**) and its **total derivative** is

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}.$$

Chain rule (two parameters). If $u = f(x, y)$ with $x = \phi(s, t)$ and $y = \psi(s, t)$, then

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s}, \qquad \frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial t}.$$

Example. $u_x = 2x$, $u_y = 2y$, $\frac{dx}{dt} = 2at$, $\frac{dy}{dt} = 2a$.

$$\frac{du}{dt} = 2x(2at) + 2y(2a) = 2at^2 \cdot 2at + 2(2at)(2a) = 4a^2t^3 + 8a^2t.$$

Verification. Substituting first, $u = a^2t^4 + 4a^2t^2$, so $\frac{du}{dt} = 4a^2t^3 + 8a^2t$. Both methods agree.

Que 1.11 **Explain differentiation of implicit functions. If $x^3 + y^3 = 3axy$, find $\frac{dy}{dx}$.**

ANSWER

If y is defined as a function of x by an equation $f(x, y) = 0$, then the total differential of f is zero along the curve:

$$f_x\,dx + f_y\,dy = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{f_x}{f_y}, \qquad f_y \neq 0.$$

Example. $f(x, y) = x^3 + y^3 - 3axy$:

$$f_x = 3x^2 - 3ay, \qquad f_y = 3y^2 - 3ax.$$

$$\frac{dy}{dx} = -\frac{3x^2 - 3ay}{3y^2 - 3ax} = \frac{ay - x^2}{y^2 - ax}.$$

NEXT ENGINEER sheerat

Que 1.12 If $z = f(x, y)$ where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, prove that $\dfrac{\partial z}{\partial u} - \dfrac{\partial z}{\partial v} = x\dfrac{\partial z}{\partial x} - y\dfrac{\partial z}{\partial y}$.

ANSWER

$$\frac{\partial x}{\partial u} = e^u, \quad \frac{\partial y}{\partial u} = -e^{-u}, \quad \frac{\partial x}{\partial v} = -e^{-v}, \quad \frac{\partial y}{\partial v} = -e^v.$$

By the chain rule,

$$\frac{\partial z}{\partial u} = z_x e^u - z_y e^{-u}, \qquad \frac{\partial z}{\partial v} = -z_x e^{-v} - z_y e^v.$$

Subtracting,

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = z_x\left(e^u + e^{-v}\right) - z_y\left(e^{-u} - e^v\right) = xz_x - yz_y. \blacksquare$$

Que 1.13 If $u = f(x - y,\ y - z,\ z - x)$, show that $u_x + u_y + u_z = 0$.

ANSWER

Put $p = x - y$, $q = y - z$, $r = z - x$, so $u = f(p, q, r)$. By the chain rule,

$$u_x = f_p \cdot 1 + f_q \cdot 0 + f_r \cdot (-1) = f_p - f_r,$$

$$u_y = f_p \cdot (-1) + f_q \cdot 1 + f_r \cdot 0 = -f_p + f_q,$$

$$u_z = f_p \cdot 0 + f_q \cdot (-1) + f_r \cdot 1 = -f_q + f_r.$$

Adding, $u_x + u_y + u_z = 0$. ∎

Que 1.14 If $z = f(x, y)$ and $x = r\cos\theta$, $y = r\sin\theta$, prove that $\left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2 = \left(\dfrac{\partial z}{\partial r}\right)^2 + \dfrac{1}{r^2}\left(\dfrac{\partial z}{\partial \theta}\right)^2$.

ANSWER

This is a **change of variables** from Cartesian to polar coordinates.

$$\frac{\partial x}{\partial r} = \cos\theta, \quad \frac{\partial y}{\partial r} = \sin\theta, \quad \frac{\partial x}{\partial \theta} = -r\sin\theta, \quad \frac{\partial y}{\partial \theta} = r\cos\theta.$$

By the chain rule,

$$z_r = z_x\cos\theta + z_y\sin\theta, \qquad z_\theta = -rz_x\sin\theta + rz_y\cos\theta.$$

So $\frac{1}{r}z_\theta = -z_x \sin\theta + z_y \cos\theta$. Squaring and adding,

$$z_r^2 + \frac{1}{r^2}z_\theta^2 = z_x^2(\cos^2\theta + \sin^2\theta) + z_y^2(\sin^2\theta + \cos^2\theta) + 2z_x z_y(\cos\theta\sin\theta - \sin\theta\cos\theta) = z_x^2 + z_y^2. \blacksquare$$

Que 1.15 **If $u = f(r)$ where $r^2 = x^2 + y^2$, prove that $u_{xx} + u_{yy} = f''(r) + \frac{1}{r}f'(r)$.**

ANSWER

From $r^2 = x^2 + y^2$, $\frac{\partial r}{\partial x} = \frac{x}{r}$ and $\frac{\partial r}{\partial y} = \frac{y}{r}$.

$$u_x = f'(r)\frac{x}{r}.$$

$$u_{xx} = f''(r)\frac{x^2}{r^2} + f'(r)\frac{\partial}{\partial x}\left(\frac{x}{r}\right) = f''(r)\frac{x^2}{r^2} + f'(r)\frac{r - x\cdot\frac{x}{r}}{r^2} = f''(r)\frac{x^2}{r^2} + f'(r)\frac{y^2}{r^3}.$$

Similarly

$$u_{yy} = f''(r)\frac{y^2}{r^2} + f'(r)\frac{x^2}{r^3}.$$

Adding and using $x^2 + y^2 = r^2$,

$$u_{xx} + u_{yy} = f''(r) + f'(r)\frac{r^2}{r^3} = f''(r) + \frac{1}{r}f'(r). \blacksquare$$

Part 4: Jacobians

Que 1.16 **Define the Jacobian and state its important properties. Find $\frac{\partial(x,y)}{\partial(r,\theta)}$ for polar coordinates and $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$ for spherical coordinates.**

ANSWER

If u and v are functions of x and y, the **Jacobian** of u, v with respect to x, y is the determinant

$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}.$$

For three functions u, v, w of x, y, z,

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}.$$

The Jacobian measures how a small area (or volume) is magnified by the change of variables: $du\,dv = |J|\,dx\,dy$.

Properties

1. $JJ' = 1$: if $J = \dfrac{\partial(u,v)}{\partial(x,y)}$ and $J' = \dfrac{\partial(x,y)}{\partial(u,v)}$, then $JJ' = 1$.
2. **Chain rule:** if u, v are functions of r, s, and r, s are functions of x, y, then $\dfrac{\partial(u,v)}{\partial(x,y)} = \dfrac{\partial(u,v)}{\partial(r,s)} \cdot \dfrac{\partial(r,s)}{\partial(x,y)}$.
3. **Functional dependence:** u and v are functionally dependent (one can be written as a function of the other, $F(u,v) = 0$) if and only if $\dfrac{\partial(u,v)}{\partial(x,y)} = 0$ identically.

Polar coordinates $x = r\cos\theta$, $y = r\sin\theta$:

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r.$$

Spherical coordinates $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$:

$$\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix}.$$

Expanding along the third row:

$$= \cos\theta\left(r^2\sin\theta\cos\theta\cos^2\phi + r^2\sin\theta\cos\theta\sin^2\phi\right) + r\sin\theta\left(r\sin^2\theta\cos^2\phi + r\sin^2\theta\sin^2\phi\right)$$

$$= r^2\sin\theta\cos^2\theta + r^2\sin^3\theta = r^2\sin\theta.$$

These two results are used for changing variables in double and triple integrals (Unit 2).

Que 1.17 **If $x = u(1-v)$ and $y = uv$, find $J = \dfrac{\partial(x,y)}{\partial(u,v)}$ and $J' = \dfrac{\partial(u,v)}{\partial(x,y)}$, and verify that $JJ' = 1$.**

ANSWER

$$J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1-v & -u \\ v & u \end{vmatrix} = u(1-v) + uv = u.$$

To find J', express u, v in terms of x, y. Adding the equations, $x + y = u$, and then $v = \dfrac{y}{u} = \dfrac{y}{x+y}$.

$$u_x = 1, \quad u_y = 1, \quad v_x = -\frac{y}{(x+y)^2}, \quad v_y = \frac{(x+y)-y}{(x+y)^2} = \frac{x}{(x+y)^2}.$$

$$J' = \begin{vmatrix} 1 & 1 \\ -\frac{y}{(x+y)^2} & \frac{x}{(x+y)^2} \end{vmatrix} = \frac{x}{(x+y)^2} + \frac{y}{(x+y)^2} = \frac{1}{x+y} = \frac{1}{u}.$$

Hence $JJ' = u \cdot \dfrac{1}{u} = 1$. ∎

Que 1.18 **Show that $u = \dfrac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$ are functionally dependent, and find the relation between them.**

ANSWER

$$u_x = \frac{(1-xy)+y(x+y)}{(1-xy)^2} = \frac{1+y^2}{(1-xy)^2}, \qquad u_y = \frac{1+x^2}{(1-xy)^2}.$$

$$v_x = \frac{1}{1+x^2}, \qquad v_y = \frac{1}{1+y^2}.$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{1+y^2}{(1-xy)^2}\cdot\frac{1}{1+y^2} - \frac{1+x^2}{(1-xy)^2}\cdot\frac{1}{1+x^2} = \frac{1}{(1-xy)^2} - \frac{1}{(1-xy)^2} = 0.$$

The Jacobian is zero, so u and v are **functionally dependent**. The relation follows from the addition formula:

$$v = \tan^{-1} x + \tan^{-1} y = \tan^{-1}\frac{x+y}{1-xy} = \tan^{-1} u, \quad \text{that is} \quad u = \tan v.$$

Part 5: Maxima and Minima

Que 1.19 **Explain the conditions for maxima and minima of a function of two variables.**

ANSWER

$f(x,y)$ has a **maximum** at (a,b) if $f(a,b) \geq f(x,y)$ for all points near (a,b), and a **minimum** if $f(a,b) \leq f(x,y)$ for all points near it. Maxima and minima together are called **extreme values**.

Necessary condition. At an extreme point,

$$f_x = 0 \quad \text{and} \quad f_y = 0.$$

Points where both partial derivatives vanish are called **stationary (critical) points**.

Sufficient condition (second derivative test). At a stationary point (a,b) let

$$r = f_{xx}(a,b), \qquad s = f_{xy}(a,b), \qquad t = f_{yy}(a,b).$$

Condition	Nature of the point
$rt - s^2 > 0$ and $r < 0$	maximum
$rt - s^2 > 0$ and $r > 0$	minimum

Condition	Nature of the point
$rt - s^2 < 0$	saddle point (neither maximum nor minimum)
$rt - s^2 = 0$	doubtful; further investigation needed

A **saddle point** is a stationary point where the surface rises in one direction and falls in another, like a horse saddle.

Working rule

1. Find f_x and f_y, solve $f_x = 0$, $f_y = 0$ to get the stationary points.
2. Find r, s, t at each point.
3. Apply the table and compute the extreme values.

Que 1.20 **Find the maximum and minimum values of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.**

ANSWER

$$f_x = 3x^2 + 3y^2 - 30x + 72, \qquad f_y = 6xy - 30y = 6y(x - 5).$$

Stationary points. From $f_y = 0$: $y = 0$ or $x = 5$.

- If $y = 0$: $3x^2 - 30x + 72 = 0 \Rightarrow x^2 - 10x + 24 = 0 \Rightarrow x = 4$ or $x = 6$.
- If $x = 5$: $75 + 3y^2 - 150 + 72 = 0 \Rightarrow 3y^2 = 3 \Rightarrow y = \pm 1$.

Stationary points: $(4, 0)$, $(6, 0)$, $(5, 1)$, $(5, -1)$.

Second derivatives. $r = 6x - 30$, $s = 6y$, $t = 6x - 30$.

Point	r	s	t	$rt - s^2$	Nature
$(4, 0)$	-6	0	-6	$36 > 0$, $r < 0$	maximum
$(6, 0)$	6	0	6	$36 > 0$, $r > 0$	minimum
$(5, 1)$	0	6	0	$-36 < 0$	saddle point
$(5, -1)$	0	-6	0	$-36 < 0$	saddle point

Values.

$$f(4, 0) = 64 - 240 + 288 = 112 \quad (\text{maximum value}),$$

$$f(6, 0) = 216 - 540 + 432 = 108 \quad (\text{minimum value}).$$

Que 1.21 **Find the stationary points of $f(x, y) = x^3 + y^3 - 3xy$ and classify them.**

ANSWER

$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$, and $f_y = 3y^2 - 3x = 0 \Rightarrow x = y^2$.

Substituting, $x = x^4 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0$ or $x = 1$. The stationary points are $(0, 0)$ and $(1, 1)$.

$r = 6x$, $s = -3$, $t = 6y$.

- At $(0, 0)$: $rt - s^2 = 0 - 9 = -9 < 0$ → **saddle point**.
- At $(1, 1)$: $rt - s^2 = 36 - 9 = 27 > 0$ and $r = 6 > 0$ → **minimum**, with minimum value $f(1, 1) = 1 + 1 - 3 = -1$.

Part 6: Lagrange's Method of Undetermined Multipliers

Que 1.22 **Explain Lagrange's method of undetermined multipliers.**

ANSWER

Often we must find the extreme values of a function **subject to a condition (constraint)**. For example, minimise the surface area of a box whose volume is fixed. Lagrange's method handles this without solving the constraint for one variable.

To find the stationary values of $u = f(x, y, z)$ subject to $\phi(x, y, z) = 0$:

1. Form the **Lagrange function**

$$F(x, y, z, \lambda) = f(x, y, z) + \lambda\,\phi(x, y, z),$$

where λ is an undetermined constant (the **Lagrange multiplier**).

2. Solve the four equations

$$F_x = f_x + \lambda\phi_x = 0, \quad F_y = f_y + \lambda\phi_y = 0, \quad F_z = f_z + \lambda\phi_z = 0, \quad \phi(x, y, z) = 0.$$

3. Eliminating λ gives the stationary points. Whether a point gives a maximum or minimum is usually decided from the physical nature of the problem.

Why it works. At a constrained extreme point, the gradient of f must be perpendicular to the constraint surface, so it must be parallel to the normal $\nabla\phi$: $\nabla f = -\lambda\nabla\phi$. This is exactly the set of equations above.

Que 1.23 **A rectangular box open at the top is to have a volume of 32 cubic units. Find the dimensions that require the least material.**

ANSWER

Let the length, breadth and height be x, y, z. The box is open at the top, so the surface area is

$$S = xy + 2xz + 2yz,$$

subject to $\phi = xyz - 32 = 0$.

$$F = xy + 2xz + 2yz + \lambda(xyz - 32).$$

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$$F_x = y + 2z + \lambda yz = 0 \tag{1}$$

$$F_y = x + 2z + \lambda xz = 0 \tag{2}$$

$$F_z = 2x + 2y + \lambda xy = 0 \tag{3}$$

Multiply (1) by x, (2) by y, (3) by z:

$$xy + 2xz = -\lambda xyz, \qquad xy + 2yz = -\lambda xyz, \qquad 2xz + 2yz = -\lambda xyz.$$

From the first two, $2xz = 2yz \Rightarrow x = y$. From the second and third, $xy = 2xz \Rightarrow y = 2z$.

So $x = y = 2z$. Using $xyz = 32$: $2z \cdot 2z \cdot z = 4z^3 = 32 \Rightarrow z = 2$, and $x = y = 4$.

Dimensions: $4 \times 4 \times 2$, and the least surface area is $S = 16 + 16 + 16 = 48$ square units.

Que 1.24 **Find the point on the plane $2x + 3y + z = 14$ nearest to the origin, and the least distance.**

ANSWER

Minimise the square of the distance, $f = x^2 + y^2 + z^2$, subject to $\phi = 2x + 3y + z - 14 = 0$.

$$F = x^2 + y^2 + z^2 + \lambda(2x + 3y + z - 14).$$

$$F_x = 2x + 2\lambda = 0, \quad F_y = 2y + 3\lambda = 0, \quad F_z = 2z + \lambda = 0.$$

So $x = -\lambda$, $y = -\frac{3}{2}\lambda$, $z = -\frac{1}{2}\lambda$. Substituting in the constraint:

$$-2\lambda - \frac{9}{2}\lambda - \frac{1}{2}\lambda = 14 \;\Rightarrow\; -7\lambda = 14 \;\Rightarrow\; \lambda = -2.$$

The point is $(2, 3, 1)$, and the least distance is $\sqrt{4 + 9 + 1} = \sqrt{14}$.

(Check: the perpendicular distance from the origin to the plane is $\dfrac{14}{\sqrt{4 + 9 + 1}} = \sqrt{14}$.)

Part 7: Errors and Approximations

Que 1.25 **Explain absolute, relative and percentage errors using differentials.**

ANSWER

If $u = f(x, y, z)$ and the measured values of x, y, z have small errors $\delta x, \delta y, \delta z$, then the error in u is approximately the total differential:

$$\delta u \approx \frac{\partial u}{\partial x}\delta x + \frac{\partial u}{\partial y}\delta y + \frac{\partial u}{\partial z}\delta z.$$

- **Absolute error:** δu.
- **Relative error:** $\dfrac{\delta u}{u}$.
- **Percentage error:** $\dfrac{\delta u}{u} \times 100$.

Useful shortcut (logarithmic differentiation). For a product of powers $u = k\,x^a y^b z^c$, take logs: $\log u = \log k + a\log x + b\log y + c\log z$. Then

$$\frac{\delta u}{u} \times 100 = a\left(\frac{\delta x}{x} \times 100\right) + b\left(\frac{\delta y}{y} \times 100\right) + c\left(\frac{\delta z}{z} \times 100\right).$$

So the percentage errors simply add up, multiplied by the powers.

Approximate calculation. $f(x+h, y+k) \approx f(x,y) + hf_x + kf_y$.

Que 1.26 **(a) The volume of a cylinder is $V = \pi r^2 h$. If the radius is measured with an error of 1% and the height with an error of 2%, find the maximum percentage error in V. (b) Find the approximate value of $(3.98)^2(2.01)^3$.**

ANSWER

(a) $\log V = \log \pi + 2\log r + \log h$, so

$$\frac{\delta V}{V} \times 100 = 2\left(\frac{\delta r}{r} \times 100\right) + \frac{\delta h}{h} \times 100 = 2(1) + 2 = 4\%.$$

The maximum percentage error in the volume is **4%** (taking both errors in the same direction).

(b) Let $f(x,y) = x^2y^3$, with $x = 4$, $\delta x = -0.02$, $y = 2$, $\delta y = 0.01$.

$$f(4,2) = 16 \times 8 = 128, \qquad f_x = 2xy^3 = 64, \qquad f_y = 3x^2y^2 = 192.$$

$$\delta f \approx 64(-0.02) + 192(0.01) = -1.28 + 1.92 = 0.64.$$

So $(3.98)^2(2.01)^3 \approx 128 + 0.64 = \mathbf{128.64}$. (The exact value is about 128.63.)

Que 1.27 **The period of a simple pendulum is $T = 2\pi\sqrt{l/g}$. Find the percentage error in T when l is measured 2% too large and g is taken 0.5% too small.**

ANSWER

$$\log T = \log 2\pi + \frac{1}{2}\log l - \frac{1}{2}\log g.$$

$$\frac{\delta T}{T} \times 100 = \frac{1}{2}\left(\frac{\delta l}{l} \times 100\right) - \frac{1}{2}\left(\frac{\delta g}{g} \times 100\right) = \frac{1}{2}(2) - \frac{1}{2}(-0.5) = 1 + 0.25 = 1.25\%.$$

The computed period is about **1.25% too large**.

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Short-Answer Questions (2 marks)

Q1. Define a homogeneous function with an example.

$f(x, y)$ is homogeneous of degree n if $f(tx, ty) = t^n f(x, y)$. Example: $x^2 + xy + y^2$ has degree 2.

Q2. State Euler's theorem.

If u is homogeneous of degree n in x, y, then $xu_x + yu_y = nu$.

Q3. If $u = x^2y + y^2x$, find $xu_x + yu_y$.

u is homogeneous of degree 3, so $xu_x + yu_y = 3u = 3(x^2y + y^2x)$.

Q4. Write the formula for the total derivative of $u = f(x, y)$ with x, y functions of t.

$\dfrac{du}{dt} = u_x \dfrac{dx}{dt} + u_y \dfrac{dy}{dt}$.

Q5. If $x = r\cos\theta$, $y = r\sin\theta$, find $\dfrac{\partial(r, \theta)}{\partial(x, y)}$.

Since $\dfrac{\partial(x, y)}{\partial(r, \theta)} = r$ and $JJ' = 1$, the answer is $\dfrac{1}{r}$.

Q6. When are two functions functionally dependent?

When their Jacobian with respect to the independent variables is identically zero.

Q7. Write the conditions for a minimum of $f(x, y)$.

$f_x = f_y = 0$, $rt - s^2 > 0$ and $r > 0$.

Q8. What is a saddle point?

A stationary point at which $rt - s^2 < 0$; the function is neither a maximum nor a minimum there.

Q9. If $u = \sin^{-1}(x/y) + \tan^{-1}(y/x)$, find $xu_x + yu_y$.

u is homogeneous of degree 0, so $xu_x + yu_y = 0$.

Q10. Find $\dfrac{dy}{dx}$ if $x^y + y^x = c$.

With $f = x^y + y^x - c$: $f_x = yx^{y-1} + y^x \log y$, $f_y = x^y \log x + xy^{x-1}$, so $\dfrac{dy}{dx} = -\dfrac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$.

Q11. If the radius of a sphere is measured with an error of 1%, what is the percentage error in its volume?

$V = \frac{4}{3}\pi r^3$, so the error is $3 \times 1\% = 3\%$.

Q12. State the Lagrange equations for extremising $f(x, y)$ subject to $\phi(x, y) = 0$.

$f_x + \lambda\phi_x = 0$, $f_y + \lambda\phi_y = 0$, $\phi(x, y) = 0$.

Practice Questions

1. If $u = e^{xyz}$, show that $\dfrac{\partial^3 u}{\partial x\, \partial y\, \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$.

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2. If $u = x^3 - 3xy^2$, verify that $u_{xx} + u_{yy} = 0$.
3. If $u = \log \dfrac{x^4 + y^4}{x + y}$, show that $xu_x + yu_y = 3$.
4. If $u = \cos^{-1} \dfrac{x + y}{\sqrt{x} + \sqrt{y}}$, prove that $xu_x + yu_y + \frac{1}{2}\cot u = 0$.
5. If $u = x^2 - y^2$, $v = 2xy$ and $x = r\cos\theta$, $y = r\sin\theta$, find $\dfrac{\partial(u, v)}{\partial(r, \theta)}$. (Answer: $4r^3$.)
6. Show that $u = x + y - z$, $v = x - y + z$, $w = x^2 + y^2 + z^2 - 2yz$ are functionally dependent and find the relation. (Answer: $u^2 + v^2 = 2w$.)
7. Find the extreme values of $x^4 + y^4 - 2x^2 + 4xy - 2y^2$. (Answer: minimum -8 at $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$; $(0, 0)$ needs further investigation.)
8. Find the maximum value of $x^2y^3z^4$ subject to $x + y + z = 9$. (Answer: at $(2, 3, 4)$, value $4 \cdot 27 \cdot 256 = 27648$.)
9. Divide 24 into three parts such that the continued product of the first, the square of the second and the cube of the third is maximum. (Answer: $4, 8, 12$.)
10. In estimating the area of a triangle $\Delta = \frac{1}{2}bc\sin A$, errors of 1% in b and c are made. Find the percentage error in the area when A is exact. (Answer: 2%.)

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UNIT 2

Multiple Integrals

Unit overview

A single integral adds up a function along a line and gives the area under a curve. A **double integral** adds up a function over a region of the plane, and a **triple integral** over a region of space. With them we find areas, volumes, masses and centres of gravity of real objects. The unit also introduces the **Beta and Gamma functions**, which turn many difficult definite integrals into simple products of factorials.

Topics in this unit: double integrals in Cartesian and polar coordinates, triple integrals, change of order of integration, change of variables, Beta and Gamma functions and their properties, Dirichlet's integral and its applications, area and volume.

Most asked: change of order of integration, change to polar coordinates, area by double integration, volume by triple integration, Beta–Gamma evaluations, and Dirichlet's integral.

Part 1: Double Integrals

Que 2.1 **Define a double integral and explain how it is evaluated as a repeated integral. Evaluate $\iint_R xy\,dx\,dy$ where R is the region bounded by $y = x$ and $y = x^2$.**

ANSWER

Definition. Let $f(x, y)$ be defined on a closed bounded region R of the xy-plane. Divide R into n small pieces of areas $\delta A_1, \ldots, \delta A_n$, and pick a point (x_i, y_i) in each. The **double integral** is

$$\iint_R f(x,y)\,dA = \lim_{n\to\infty,\,\delta A_i\to 0} \sum_{i=1}^{n} f(x_i, y_i)\,\delta A_i.$$

If $f(x, y) = 1$, the double integral gives the **area** of R. If $f(x, y) \geq 0$, it gives the **volume** of the solid under the surface $z = f(x, y)$ above R.

Evaluation as a repeated integral.

- If R is bounded by $x = a$, $x = b$ and the curves $y = y_1(x)$, $y = y_2(x)$ (vertical strips):

$$\iint_R f\,dA = \int_a^b \left[\int_{y_1(x)}^{y_2(x)} f(x,y)\,dy\right] dx.$$

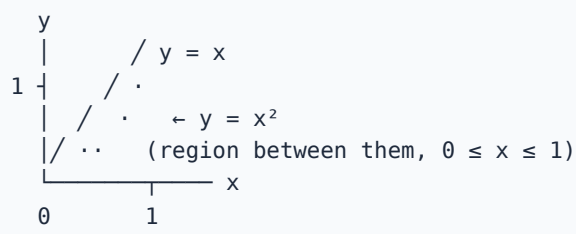
Integrate first with respect to y, treating x as constant.

- If R is bounded by $y = c$, $y = d$ and the curves $x = x_1(y)$, $x = x_2(y)$ (horizontal strips):

$$\iint_R f\,dA = \int_c^d \left[\int_{x_1(y)}^{x_2(y)} f(x,y)\,dx\right] dy.$$

Rule: the inner integral has variable limits and is evaluated first; the outer integral always has constant limits.

Example. The curves $y = x$ and $y = x^2$ meet where $x^2 = x$, that is at $x = 0$ and $x = 1$. For $0 \le x \le 1$, the parabola lies below the line, so a vertical strip goes from $y = x^2$ to $y = x$.



$$\iint_R xy\,dx\,dy = \int_0^1 x\left[\frac{y^2}{2}\right]_{x^2}^{x} dx = \frac{1}{2}\int_0^1 x\left(x^2 - x^4\right) dx = \frac{1}{2}\left(\frac{1}{4} - \frac{1}{6}\right) = \frac{1}{24}.$$

Que 2.2 Evaluate $\int_0^1 \int_0^{2x} dy\,dx$ and interpret the result. Also evaluate $\int_0^1 \int_0^2 (x^2 + y^2)\,dy\,dx$.

ANSWER

(a) $\int_0^1 \int_0^{2x} dy\,dx = \int_0^1 2x\,dx = 1.$

This is the area of the triangle bounded by $y = 0$, $x = 1$ and $y = 2x$, which has base 1 and height 2, so its area is $\frac{1}{2} \cdot 1 \cdot 2 = 1$. ✓

(b) Integrate first with respect to y:

$$\int_0^2 (x^2 + y^2)\,dy = \left[x^2 y + \frac{y^3}{3}\right]_0^2 = 2x^2 + \frac{8}{3}.$$

$$\int_0^1 \left(2x^2 + \frac{8}{3}\right) dx = \frac{2}{3} + \frac{8}{3} = \frac{10}{3}.$$

Part 2: Change of Order of Integration

Que 2.3 Explain the method of changing the order of integration. Change the order and evaluate $\int_0^a \int_x^a \frac{x}{x^2 + y^2}\,dy\,dx$.

ANSWER

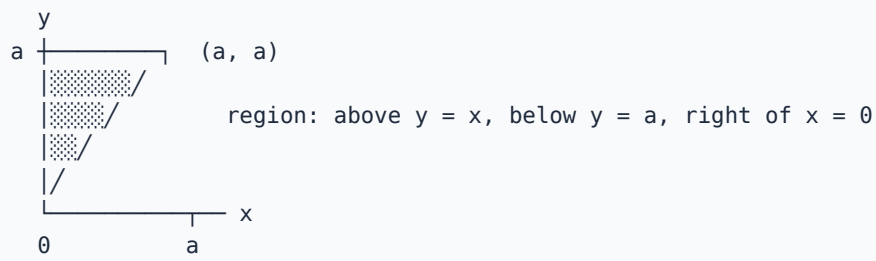
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Sometimes the inner integral is hard or impossible in the given order but easy in the other order. Changing the order means describing the **same region** with strips in the other direction.

Method

1. From the given limits, write the four boundaries of the region and **sketch** it.
2. If the given strips are vertical (inner variable y), draw horizontal strips instead (inner variable x), or the other way round.
3. Read off the new limits: the inner limits are the curves where a strip enters and leaves the region, written in terms of the outer variable; the outer limits are the extreme values of the outer variable.
4. If one family of strips does not cover the region with a single formula, split the region into parts.

Example. The given limits are $x \le y \le a$, $0 \le x \le a$. The region is the triangle bounded by $y = x$, $y = a$ and $x = 0$, with vertices $(0, 0)$, $(0, a)$, (a, a).



In the given order, $\int \frac{x}{x^2+y^2} dy = \tan^{-1}(y/x)$, which leads to a hard outer integral. Use **horizontal strips**: for a fixed y between 0 and a, x goes from 0 to y.

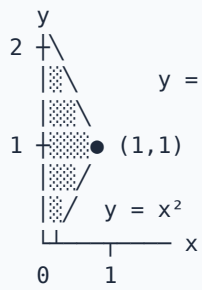
$$I = \int_0^a \int_0^y \frac{x}{x^2+y^2}\, dx\, dy = \int_0^a \left[\frac{1}{2}\log(x^2+y^2)\right]_0^y dy = \int_0^a \frac{1}{2}\log\frac{2y^2}{y^2}\, dy = \frac{1}{2}\log 2 \int_0^a dy.$$

$$I = \frac{a}{2}\log 2.$$

Que 2.4 **Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$ and evaluate it both ways.**

ANSWER

Region. $0 \le x \le 1$ and $x^2 \le y \le 2 - x$. It is bounded by the parabola $y = x^2$, the line $y = 2 - x$ and the y-axis. The parabola and line meet at $x^2 = 2 - x$, that is $x = 1$, $y = 1$.



Given order:

$$\int_0^1 x\left[\frac{y^2}{2}\right]_{x^2}^{2-x} dx = \frac{1}{2}\int_0^1 x\left[(2-x)^2 - x^4\right] dx = \frac{1}{2}\int_0^1 \left(4x - 4x^2 + x^3 - x^5\right) dx$$

$$= \frac{1}{2}\left(2 - \frac{4}{3} + \frac{1}{4} - \frac{1}{6}\right) = \frac{1}{2}\cdot\frac{24 - 16 + 3 - 2}{12} = \frac{9}{24} = \frac{3}{8}.$$

Changed order. A horizontal strip at height y starts at $x = 0$. It ends on the parabola $x = \sqrt{y}$ when $0 \le y \le 1$, and on the line $x = 2 - y$ when $1 \le y \le 2$. So the region splits into two parts:

$$I = \int_0^1 \int_0^{\sqrt{y}} xy\, dx\, dy + \int_1^2 \int_0^{2-y} xy\, dx\, dy.$$

$$\text{First part} = \int_0^1 y \cdot \frac{y}{2}\, dy = \frac{1}{6}.$$

$$\text{Second part} = \int_1^2 y \cdot \frac{(2-y)^2}{2}\, dy = \frac{1}{2}\int_1^2 (4y - 4y^2 + y^3)\, dy = \frac{1}{2}\left(6 - \frac{28}{3} + \frac{15}{4}\right) = \frac{5}{24}.$$

$$I = \frac{1}{6} + \frac{5}{24} = \frac{9}{24} = \frac{3}{8}. \checkmark$$

Que 2.5 By changing the order of integration, evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\, dy\, dx.$

ANSWER

The inner integral $\int \frac{e^{-y}}{y} dy$ cannot be found in elementary terms, so the order must be changed.

Region: $0 \le x < \infty$, $x \le y < \infty$, that is the part of the first quadrant above the line $y = x$. With horizontal strips, for each $y \ge 0$, x runs from 0 to y:

$$I = \int_0^\infty \int_0^y \frac{e^{-y}}{y}\, dx\, dy = \int_0^\infty \frac{e^{-y}}{y} \cdot y\, dy = \int_0^\infty e^{-y}\, dy = 1.$$

NEXT ENGINEER _sheerat_

Part 3: Change of Variables and Polar Coordinates

Que 2.6 **Explain the change of variables in a double integral. Write the formula for changing to polar coordinates.**

ANSWER

If $x = x(u,v)$ and $y = y(u,v)$ map a region R' of the uv-plane one-to-one onto the region R of the xy-plane, then

$$\iint_R f(x,y)\,dx\,dy = \iint_{R'} f\big(x(u,v),\,y(u,v)\big)\,|J|\,du\,dv, \qquad J = \frac{\partial(x,y)}{\partial(u,v)}.$$

The Jacobian (Unit 1) accounts for the stretching of area elements.

Polar coordinates. With $x = r\cos\theta$, $y = r\sin\theta$, $J = r$, so

$$dx\,dy = r\,dr\,d\theta, \qquad \iint_R f(x,y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta.$$

Polar coordinates are useful whenever the region is a circle, a sector, or is bounded by curves like $r = a(1+\cos\theta)$, and whenever the integrand contains x^2+y^2.

Que 2.7 **By changing to polar coordinates, evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dx\,dy$. Hence show that $\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$.**

ANSWER

The region is the whole first quadrant: $0 \le r < \infty$, $0 \le \theta \le \frac{\pi}{2}$, and $x^2+y^2 = r^2$.

$$I = \int_0^{\pi/2} \int_0^\infty e^{-r^2}\,r\,dr\,d\theta.$$

Put $r^2 = t$, $2r\,dr = dt$:

$$\int_0^\infty e^{-r^2}\,r\,dr = \frac{1}{2}\int_0^\infty e^{-t}dt = \frac{1}{2}.$$

$$I = \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4}.$$

Deduction. Since the variables separate,

$$I = \left(\int_0^\infty e^{-x^2}dx\right)\left(\int_0^\infty e^{-y^2}dy\right) = \left(\int_0^\infty e^{-x^2}dx\right)^2 = \frac{\pi}{4}.$$

Hence $\int_0^\infty e^{-x^2}\,dx = \dfrac{\sqrt{\pi}}{2}$. This is the **Gaussian integral**, central to probability and statistics.

Que 2.8 **Evaluate $\iint_R (x+y)^2\,dx\,dy$, where R is the parallelogram with vertices $(1,0)$, $(3,1)$, $(2,2)$, $(0,1)$, using the substitution $u = x+y$, $v = x-2y$.**

ANSWER

New limits. The sides of the parallelogram are:

Side	Equation	In new variables
$(1,0)$ to $(0,1)$	$x+y=1$	$u=1$
$(3,1)$ to $(2,2)$	$x+y=4$	$u=4$
$(1,0)$ to $(3,1)$	$x-2y=1$	$v=1$
$(0,1)$ to $(2,2)$	$x-2y=-2$	$v=-2$

So the parallelogram becomes the rectangle $1 \le u \le 4$, $-2 \le v \le 1$.

Jacobian.

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = -3 \quad \Rightarrow \quad \frac{\partial(x,y)}{\partial(u,v)} = -\frac{1}{3}, \quad |J| = \frac{1}{3}.$$

Integral.

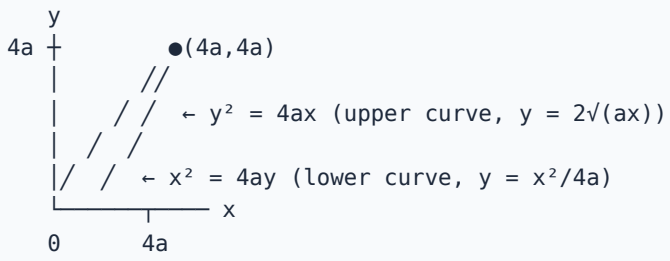
$$\iint_R (x+y)^2\,dx\,dy = \int_{-2}^{1}\int_{1}^{4} u^2 \cdot \frac{1}{3}\,du\,dv = \frac{1}{3}\cdot\left[\frac{u^3}{3}\right]_1^4 \cdot 3 = \frac{1}{3}\cdot 21 \cdot 3 = 21.$$

Part 4: Area by Double Integration

Que 2.9 **Find, by double integration, the area enclosed between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.**

ANSWER

Points of intersection. From $x^2 = 4ay$, $y = \dfrac{x^2}{4a}$. Substituting in $y^2 = 4ax$: $\dfrac{x^4}{16a^2} = 4ax \Rightarrow x(x^3 - 64a^3) = 0 \Rightarrow x = 0$ or $x = 4a$. The curves meet at $(0,0)$ and $(4a, 4a)$.



A vertical strip runs from $y = \dfrac{x^2}{4a}$ up to $y = 2\sqrt{ax}$:

$$A = \int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy\,dx = \int_0^{4a} \left(2\sqrt{a}\,x^{1/2} - \frac{x^2}{4a}\right) dx = \left[\frac{4\sqrt{a}}{3}x^{3/2} - \frac{x^3}{12a}\right]_0^{4a}.$$

$$= \frac{4\sqrt{a}}{3}\cdot 8a^{3/2} - \frac{64a^3}{12a} = \frac{32a^2}{3} - \frac{16a^2}{3} = \frac{16a^2}{3}.$$

Que 2.10 **Find the area of the cardioid $r = a(1+\cos\theta)$.**

ANSWER

In polar coordinates the area element is $r\,dr\,d\theta$. The cardioid is symmetric about the initial line, and θ goes from 0 to 2π (or twice from 0 to π).

$$A = \int_0^{2\pi} \int_0^{a(1+\cos\theta)} r\,dr\,d\theta = \int_0^{2\pi} \frac{a^2(1+\cos\theta)^2}{2}\,d\theta = \frac{a^2}{2}\int_0^{2\pi} \left(1 + 2\cos\theta + \cos^2\theta\right) d\theta.$$

Using $\int_0^{2\pi}\cos\theta\,d\theta = 0$ and $\int_0^{2\pi}\cos^2\theta\,d\theta = \pi$:

$$A = \frac{a^2}{2}(2\pi + 0 + \pi) = \frac{3\pi a^2}{2}.$$

Part 5: Triple Integrals and Volume

Que 2.11 **Define a triple integral. Evaluate (a) $\displaystyle\int_0^1\int_0^2\int_0^3 xyz^2\,dz\,dy\,dx$ and (b) $\displaystyle\iiint_V xyz\,dx\,dy\,dz$ over the tetrahedron bounded by $x=0$, $y=0$, $z=0$ and $x+y+z=1$.**

ANSWER

Definition. For $f(x,y,z)$ defined on a region V of space, divide V into small volumes δV_i and form $\sum f(x_i,y_i,z_i)\,\delta V_i$. The limit as the pieces shrink is the **triple integral** $\iiint_V f\,dV$. With $f = 1$ it gives the **volume** of V; with f = density, it gives the **mass**. It is evaluated as three repeated integrals, innermost first.

(a) The limits are constants, so the integral separates:

$$\int_0^1 x\,dx \cdot \int_0^2 y\,dy \cdot \int_0^3 z^2\,dz = \frac{1}{2}\cdot 2 \cdot 9 = 9.$$

(b) For a point (x, y) in the base triangle, z runs from 0 to $1 - x - y$; then y from 0 to $1 - x$, and x from 0 to 1.

$$I = \int_0^1 \int_0^{1-x} xy\,\frac{(1-x-y)^2}{2}\,dy\,dx.$$

Put $c = 1 - x$. Then

$$\int_0^c y(c-y)^2\,dy = \int_0^c \left(c^2 y - 2cy^2 + y^3\right) dy = \frac{c^4}{2} - \frac{2c^4}{3} + \frac{c^4}{4} = \frac{c^4}{12}.$$

$$I = \frac{1}{2}\int_0^1 x \cdot \frac{(1-x)^4}{12}\,dx = \frac{1}{24}\int_0^1 x(1-x)^4\,dx = \frac{1}{24}\cdot B(2,5) = \frac{1}{24}\cdot\frac{1!\,4!}{6!} = \frac{1}{24}\cdot\frac{1}{30} = \frac{1}{720}.$$

(The same answer follows at once from Dirichlet's integral, Que 2.18.)

Que 2.12 **Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = 0$ and $y + z = 4$.**

ANSWER

The solid stands on the disc $x^2 + y^2 \le 4$ in the xy-plane, and for each point of the disc z goes from 0 up to the plane $z = 4 - y$ (which is positive on the disc, since $y \le 2$).

$$V = \iint_{x^2+y^2\le 4} (4 - y)\,dx\,dy.$$

In polar coordinates ($y = r\sin\theta$):

$$V = \int_0^{2\pi}\int_0^2 (4 - r\sin\theta)\,r\,dr\,d\theta = \int_0^{2\pi}\left(8 - \frac{8}{3}\sin\theta\right)d\theta = 16\pi - 0 = 16\pi.$$

Que 2.13 **Using spherical coordinates, find the volume of a sphere of radius a, and evaluate $\iiint \frac{dx\,dy\,dz}{\sqrt{x^2+y^2+z^2}}$ over the same sphere.**

ANSWER

In spherical coordinates $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$, and (Unit 1, Que 1.16) $dx\,dy\,dz = r^2\sin\theta\,dr\,d\theta\,d\phi$. The sphere is $0 \le r \le a$, $0 \le \theta \le \pi$, $0 \le \phi \le 2\pi$.

Volume.

NEXT ENGINEER _sheerat_

$$V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2 \sin\theta \, dr \, d\theta \, d\phi = \frac{a^3}{3} \cdot [-\cos\theta]_0^{\pi} \cdot 2\pi = \frac{a^3}{3} \cdot 2 \cdot 2\pi = \frac{4}{3}\pi a^3.$$

Integral. Here $\sqrt{x^2+y^2+z^2} = r$:

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{a} \frac{1}{r} r^2 \sin\theta \, dr \, d\theta \, d\phi = \frac{a^2}{2} \cdot 2 \cdot 2\pi = 2\pi a^2.$$

Part 6: Beta and Gamma Functions

Que 2.14 **Define the Beta and Gamma functions and state their important properties.**

ANSWER

Gamma function. For $n > 0$,

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} \, dx.$$

Beta function. For $m > 0$, $n > 0$,

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} \, dx.$$

Properties of the Gamma function

1. $\Gamma(1) = 1$.
2. **Recurrence:** $\Gamma(n+1) = n\,\Gamma(n)$. (Proof: integrate $\int_0^{\infty} e^{-x}x^n dx$ by parts.)
3. For a positive integer n: $\Gamma(n+1) = n!$. So Γ extends the factorial.
4. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. (Put $x = t^2$: $\Gamma\left(\frac{1}{2}\right) = 2\int_0^{\infty} e^{-t^2} dt = 2 \cdot \frac{\sqrt{\pi}}{2}$, using Que 2.7.)
5. For negative non-integers, define $\Gamma(n) = \dfrac{\Gamma(n+1)}{n}$. Example: $\Gamma\left(-\frac{1}{2}\right) = \dfrac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} = -2\sqrt{\pi}$.
6. **Reflection formula:** $\Gamma(n)\Gamma(1-n) = \dfrac{\pi}{\sin n\pi}$, $0 < n < 1$.
7. **Duplication formula (Legendre):** $\Gamma(m)\Gamma\left(m+\frac{1}{2}\right) = \dfrac{\sqrt{\pi}}{2^{2m-1}}\Gamma(2m)$.

Properties of the Beta function

1. **Symmetry:** $B(m,n) = B(n,m)$ (put $x = 1-t$).
2. **Relation with Gamma:**

$$B(m,n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}.$$

3. **Trigonometric form** (put $x = \sin^2\theta$):

NEXT ENGINEER _sheerat_

$$B(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta\,\cos^{2n-1}\theta\,d\theta,$$

which gives the very useful result

$$\int_0^{\pi/2} \sin^p\theta\,\cos^q\theta\,d\theta = \frac{1}{2}B\left(\frac{p+1}{2},\frac{q+1}{2}\right) = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}.$$

4. **Improper-integral form:** $B(m,n) = \displaystyle\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx.$

Que 2.15 **Prove that $\Gamma(n+1) = n\Gamma(n)$ and find $\Gamma\left(\frac{5}{2}\right)$ and $\Gamma\left(-\frac{3}{2}\right)$.**

ANSWER

Integrating by parts with $u = x^n$ and $dv = e^{-x}dx$:

$$\Gamma(n+1) = \int_0^\infty e^{-x}x^n\,dx = \left[-x^n e^{-x}\right]_0^\infty + n\int_0^\infty e^{-x}x^{n-1}\,dx = 0 + n\Gamma(n).$$

(The bracket vanishes at both limits for $n > 0$.)

$$\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3\sqrt{\pi}}{4}.$$

$$\Gamma\left(-\frac{3}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}\right)}{-\frac{3}{2}} = \frac{-2\sqrt{\pi}}{-\frac{3}{2}} = \frac{4\sqrt{\pi}}{3}.$$

Que 2.16 **Using Beta and Gamma functions, evaluate: (a) $\displaystyle\int_0^\infty x^4e^{-x}\,dx$, (b) $\displaystyle\int_0^1 x^3(1-x)^4\,dx$, (c) $\displaystyle\int_0^{\pi/2}\sin^5\theta\cos^3\theta\,d\theta$, (d) $\displaystyle\int_0^\infty \sqrt{x}\,e^{-x^3}\,dx$.**

ANSWER

(a) $\displaystyle\int_0^\infty x^4e^{-x}dx = \Gamma(5) = 4! = 24.$

(b) This is $B(4,5)$:

$$B(4,5) = \frac{\Gamma(4)\Gamma(5)}{\Gamma(9)} = \frac{3!\,4!}{8!} = \frac{6\cdot 24}{40320} = \frac{1}{280}.$$

(c) With $p = 5$, $q = 3$:

$$\int_0^{\pi/2}\sin^5\theta\cos^3\theta\,d\theta = \frac{\Gamma(3)\,\Gamma(2)}{2\,\Gamma(5)} = \frac{2\cdot 1}{2\cdot 24} = \frac{1}{24}.$$

(d) Put $x^3 = t$, so $x = t^{1/3}$, $dx = \frac{1}{3}t^{-2/3}dt$ and $\sqrt{x} = t^{1/6}$:

$$\int_0^\infty t^{1/6} e^{-t} \cdot \frac{1}{3} t^{-2/3}\, dt = \frac{1}{3}\int_0^\infty t^{-1/2} e^{-t}\, dt = \frac{1}{3}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{3}.$$

Que 2.17 **Prove that $B(m,n) = \dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.**

ANSWER

Putting $x = t^2$ in the definition of Γ gives $\Gamma(m) = 2\int_0^\infty e^{-t^2} t^{2m-1} dt$. Hence

$$\Gamma(m)\Gamma(n) = 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2m-1} y^{2n-1}\, dx\, dy.$$

Change to polar coordinates, $x = r\cos\theta$, $y = r\sin\theta$, $dx\,dy = r\,dr\,d\theta$, over the first quadrant:

$$\Gamma(m)\Gamma(n) = 4\int_0^{\pi/2} \int_0^\infty e^{-r^2} r^{2m+2n-1} \cos^{2m-1}\theta\, \sin^{2n-1}\theta\, dr\, d\theta$$

$$= \left(2\int_0^\infty e^{-r^2} r^{2(m+n)-1} dr\right)\left(2\int_0^{\pi/2} \cos^{2m-1}\theta \sin^{2n-1}\theta\, d\theta\right) = \Gamma(m+n)\cdot B(m,n),$$

using the forms of Γ and B above. Dividing by $\Gamma(m+n)$ proves the result. ∎

Part 7: Dirichlet's Integral

Que 2.18 **State Dirichlet's integral and its extension (Liouville's form). Use it to find the volume of the tetrahedron bounded by the coordinate planes and $x + y + z = 1$, and to evaluate $\iiint xyz\, dV$ over it.**

ANSWER

Dirichlet's theorem. If V is the region $x \ge 0$, $y \ge 0$, $z \ge 0$, $x + y + z \le 1$, then

$$\iiint_V x^{l-1} y^{m-1} z^{n-1}\, dx\, dy\, dz = \frac{\Gamma(l)\,\Gamma(m)\,\Gamma(n)}{\Gamma(l+m+n+1)}.$$

General (Liouville) form. If V is $x, y, z \ge 0$ and $\left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q + \left(\frac{z}{c}\right)^r \le 1$, then

$$\iiint_V x^{l-1} y^{m-1} z^{n-1}\, dV = \frac{a^l b^m c^n}{pqr} \cdot \frac{\Gamma\left(\frac{l}{p}\right)\Gamma\left(\frac{m}{q}\right)\Gamma\left(\frac{n}{r}\right)}{\Gamma\left(\frac{l}{p}+\frac{m}{q}+\frac{n}{r}+1\right)}.$$

The two-variable form is $\displaystyle\iint_{x+y\le 1} x^{l-1}y^{m-1}\,dx\,dy = \frac{\Gamma(l)\Gamma(m)}{\Gamma(l+m+1)}$.

Volume of the tetrahedron ($l=m=n=1$):

$$V = \frac{\Gamma(1)^3}{\Gamma(4)} = \frac{1}{3!} = \frac{1}{6}.$$

$\iiint xyz\,dV$ ($l=m=n=2$):

$$\frac{\Gamma(2)^3}{\Gamma(7)} = \frac{1}{6!} = \frac{1}{720},$$

which agrees with the direct evaluation in Que 2.11.

Que 2.19 **Using Dirichlet's integral, find the volume of the ellipsoid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$, and evaluate $\iiint xyz\,dV$ over the positive octant of the sphere $x^2+y^2+z^2\le a^2$.**

ANSWER

Volume of the ellipsoid. The ellipsoid is symmetric in all eight octants, so $V = 8\times$ (volume in the positive octant). Use the Liouville form with $l=m=n=1$ and $p=q=r=2$:

$$\frac{V}{8} = \frac{abc}{8}\cdot\frac{\Gamma\left(\frac{1}{2}\right)^3}{\Gamma\left(\frac{3}{2}+1\right)} = \frac{abc}{8}\cdot\frac{\pi^{3/2}}{\Gamma\left(\frac{5}{2}\right)} = \frac{abc}{8}\cdot\frac{\pi^{3/2}}{\frac{3}{4}\sqrt{\pi}} = \frac{abc}{8}\cdot\frac{4\pi}{3}.$$

$$V = \frac{4}{3}\pi abc.$$

For $a=b=c$ this gives the volume of a sphere, $\frac{4}{3}\pi a^3$.

$\iiint xyz\,dV$ **over the octant of the sphere.** Here $l=m=n=2$, $p=q=r=2$ and $a=b=c$:

$$\frac{a^6}{8}\cdot\frac{\Gamma(1)^3}{\Gamma(1+1+1+1)} = \frac{a^6}{8}\cdot\frac{1}{3!} = \frac{a^6}{48}.$$

(Check in spherical coordinates: $\int_0^a r^5 dr\int_0^{\pi/2}\sin^3\theta\cos\theta\,d\theta\int_0^{\pi/2}\sin\phi\cos\phi\,d\phi = \frac{a^6}{6}\cdot\frac{1}{4}\cdot\frac{1}{2} = \frac{a^6}{48}$. ✓)

NEXT ENGINEER _sheerat_

Short-Answer Questions (2 marks)

Q1. Evaluate $\int_0^1 \int_0^1 (x+y)\,dx\,dy$.

$\int_0^1 \left(\frac{1}{2}+y\right) dy = \frac{1}{2}+\frac{1}{2} = 1.$

Q2. Write the area element in polar coordinates.

$dA = r\,dr\,d\theta.$

Q3. Write the volume element in spherical coordinates.

$dV = r^2 \sin\theta\,dr\,d\theta\,d\phi.$

Q4. Why do we change the order of integration?

When the inner integral cannot be evaluated (or is difficult) in the given order but becomes easy in the other order.

Q5. Find $\Gamma(6)$ and $\Gamma\left(\frac{7}{2}\right)$.

$\Gamma(6) = 5! = 120;\ \Gamma\left(\frac{7}{2}\right) = \frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{15\sqrt{\pi}}{8}.$

Q6. Write the relation between Beta and Gamma functions.

$$B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$$

Q7. Evaluate $\int_0^{\pi/2} \sin^4\theta\,d\theta$.

$$\frac{\Gamma(\frac{5}{2})\Gamma(\frac{1}{2})}{2\Gamma(3)} = \frac{\frac{3}{4}\sqrt{\pi}\cdot\sqrt{\pi}}{4} = \frac{3\pi}{16}.$$

Q8. Show that $B(m,n) = B(n,m)$.

Put $x = 1-t$ in $\int_0^1 x^{m-1}(1-x)^{n-1}dx$; it becomes $\int_0^1 t^{n-1}(1-t)^{m-1}dt = B(n,m)$.

Q9. Evaluate $\int_0^{\pi} \int_0^{a} r\,dr\,d\theta$.

$\frac{a^2}{2}\cdot\pi = \frac{\pi a^2}{2}$ (the area of a semicircle).

Q10. State Dirichlet's integral.

$$\iiint_{x+y+z\le 1} x^{l-1}y^{m-1}z^{n-1}\,dV = \frac{\Gamma(l)\Gamma(m)\Gamma(n)}{\Gamma(l+m+n+1)}.$$

Q11. Write the value of $\Gamma\left(\frac{1}{2}\right)$.

$\sqrt{\pi}.$

Q12. Find $\int_0^{\infty} e^{-4x}x^3\,dx$.

Put $4x = t$: $\frac{1}{4^4}\Gamma(4) = \frac{6}{256} = \frac{3}{128}.$

NEXT ENGINEER _sheerat_

Practice Questions

1. Evaluate $\iint_R xy\,dA$ over the positive quadrant of the circle $x^2+y^2=a^2$. (Answer: $\frac{a^4}{8}$.)
2. Change the order of integration in $\int_0^a \int_{x^2/a}^{2a-x} xy\,dy\,dx$ and evaluate. (Answer: $\frac{3a^4}{8}$.)
3. Change the order and evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$. (Answer: $\frac{e-1}{2}$.)
4. Change to polar coordinates and evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} (x^2+y^2)\,dy\,dx$. (Answer: $\frac{\pi a^4}{8}$.)
5. Find the area of one loop of the lemniscate $r^2=a^2\cos 2\theta$. (Answer: $\frac{a^2}{2}$.)
6. Find the volume of the solid bounded by the paraboloid $z=x^2+y^2$ and the plane $z=4$. (Answer: 8π.)
7. Evaluate $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$ in terms of Gamma functions. (Answer: $\frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{4\Gamma(\frac{3}{4})}$.)
8. Prove that $\int_0^{\pi/2} \sqrt{\tan\theta}\,d\theta = \frac{\pi}{\sqrt{2}}$.
9. Using Dirichlet's integral, find the mass of the octant of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1$ when the density is $\rho=kxyz$. (Answer: $\frac{ka^2b^2c^2}{48}$.)
10. Show that $\int_0^\infty \frac{x^2}{(1+x^4)}\,dx = \frac{\pi}{2\sqrt{2}}$ using Beta functions.

NEXT ENGINEER _sheerat_

UNIT 3

Vector Calculus

Unit overview

Many physical quantities vary from point to point in space. Temperature is a **scalar field** (one number at each point); the velocity of a fluid or an electric field is a **vector field** (a vector at each point). Vector calculus gives the tools to describe how such fields change (gradient, divergence, curl) and to add them up along curves and over surfaces (line, surface and volume integrals). The unit ends with the two great integral theorems of Stokes and Gauss.

Topics in this unit: gradient and its geometric meaning, directional derivative, divergence and curl, solenoidal and irrotational fields, vector identities, line integrals and work done, conservative fields, surface and volume integrals, Green's theorem in the plane, Stokes' theorem and Gauss's divergence theorem.

Most asked: directional derivative, angle between surfaces, solenoidal and irrotational conditions with constants to find, scalar potential, work done by a force, and verification or application of Stokes' and Gauss's theorems.

Part 1: Gradient and Directional Derivative

Que 3.1 **Define the vector differential operator ∇ and the gradient of a scalar field. Explain the geometric meaning of the gradient.**

ANSWER

The **vector differential operator** (del or nabla) is

$$\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}.$$

If $\phi(x, y, z)$ is a differentiable scalar field, its **gradient** is the vector field

$$\nabla\phi = \operatorname{grad}\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}.$$

Geometric meaning

1. **Normal to a level surface.** The surfaces $\phi(x, y, z) = c$ are called level surfaces. At every point, $\nabla\phi$ is **perpendicular (normal)** to the level surface through that point. So the unit normal to the surface $\phi = c$ is

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}.$$

Reason: for any curve $\vec{r}(t)$ lying on the surface, $\phi(\vec{r}(t)) = c$, so $\dfrac{d\phi}{dt} = \nabla\phi \cdot \dfrac{d\vec{r}}{dt} = 0$. Thus $\nabla\phi$ is perpendicular to every tangent direction of the surface.

2. **Direction of fastest increase.** $\nabla\phi$ points in the direction in which ϕ increases most rapidly, and $|\nabla\phi|$ is that maximum rate of increase.

Que 3.2 **Define the directional derivative. Find the directional derivative of $\phi = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$. Also find the maximum directional derivative at that point.**

ANSWER

The **directional derivative** of ϕ at a point in the direction of a unit vector $\hat{a}$ is the rate of change of ϕ per unit distance in that direction:

$$D_{\hat{a}}\phi = \nabla\phi \cdot \hat{a}.$$

Since $\nabla\phi \cdot \hat{a} = |\nabla\phi| \cos\theta$, it is greatest when $\hat{a}$ is along $\nabla\phi$; the **maximum directional derivative** is $|\nabla\phi|$.

Example.

$$\nabla\phi = y^2\,\hat{i} + (2xy + z^3)\,\hat{j} + 3yz^2\,\hat{k}.$$

At $(2, -1, 1)$:

$$\nabla\phi = 1\,\hat{i} + (-4 + 1)\,\hat{j} + (-3)\,\hat{k} = \hat{i} - 3\hat{j} - 3\hat{k}.$$

Unit vector in the given direction:

$$\hat{a} = \frac{\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{1 + 4 + 4}} = \frac{\hat{i} + 2\hat{j} + 2\hat{k}}{3}.$$

$$D_{\hat{a}}\phi = \frac{(1)(1) + (-3)(2) + (-3)(2)}{3} = \frac{1 - 6 - 6}{3} = -\frac{11}{3}.$$

The negative sign means ϕ decreases in this direction.

Maximum directional derivative $= |\nabla\phi| = \sqrt{1 + 9 + 9} = \sqrt{19}$, in the direction of $\hat{i} - 3\hat{j} - 3\hat{k}$.

Que 3.3 **Find the unit normal to the surface $x^2 + y^2 - z = 1$ at $(1, 1, 1)$, and the angle between the surfaces $x^2 + y^2 + z^2 = 6$ and $z = x^2 + y^2$ at the point $(1, 1, 2)$.**

ANSWER

(a) Let $\phi = x^2 + y^2 - z$. Then $\nabla\phi = 2x\,\hat{i} + 2y\,\hat{j} - \hat{k}$, which at $(1, 1, 1)$ is $2\hat{i} + 2\hat{j} - \hat{k}$, of magnitude $\sqrt{4 + 4 + 1} = 3$.

NEXT ENGINEER _sheerat_

$$\hat{n} = \frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}.$$

(b) The angle between two surfaces at a common point is the angle between their normals.

Check: $(1, 1, 2)$ lies on both surfaces ($1 + 1 + 4 = 6$ and $2 = 1 + 1$).

$\phi_1 = x^2 + y^2 + z^2 - 6$: $\nabla\phi_1 = 2x\,\hat{i} + 2y\,\hat{j} + 2z\,\hat{k} = 2\hat{i} + 2\hat{j} + 4\hat{k}$, $|\nabla\phi_1| = \sqrt{24} = 2\sqrt{6}$.

$\phi_2 = x^2 + y^2 - z$: $\nabla\phi_2 = 2\hat{i} + 2\hat{j} - \hat{k}$, $|\nabla\phi_2| = 3$.

$$\cos\theta = \frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1||\nabla\phi_2|} = \frac{4 + 4 - 4}{2\sqrt{6} \cdot 3} = \frac{4}{6\sqrt{6}} = \frac{2}{3\sqrt{6}}.$$

$$\theta = \cos^{-1}\left(\frac{2}{3\sqrt{6}}\right) \approx 74.2^\circ.$$

Que 3.4 **If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$, prove that $\nabla r^n = n\,r^{n-2}\,\vec{r}$. Deduce ∇r and $\nabla\left(\frac{1}{r}\right)$.**

ANSWER

From $r^2 = x^2 + y^2 + z^2$: $\dfrac{\partial r}{\partial x} = \dfrac{x}{r}$, and similarly for y and z.

$$\nabla r^n = \sum \hat{i}\,\frac{\partial}{\partial x}(r^n) = \sum \hat{i}\,nr^{n-1}\frac{x}{r} = nr^{n-2}\left(x\hat{i} + y\hat{j} + z\hat{k}\right) = nr^{n-2}\vec{r}.\ \blacksquare$$

- $n = 1$: $\nabla r = \dfrac{\vec{r}}{r} = \hat{r}$, the unit radial vector.
- $n = -1$: $\nabla\left(\dfrac{1}{r}\right) = -\dfrac{\vec{r}}{r^3}$. (This is the field of a point charge or point mass, up to a constant.)

Part 2: Divergence and Curl

Que 3.5 **Define divergence and curl of a vector field and explain their physical meaning. Define solenoidal and irrotational fields.**

ANSWER

Let $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$.

Divergence (a scalar):

$$\nabla \cdot \vec{F} = \operatorname{div}\vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

Curl (a vector):

$$\nabla \times \vec{F} = \operatorname{curl} \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\hat{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\hat{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\hat{k}.$$

Physical meaning

- If $\vec{F}$ is the velocity of a fluid, $\nabla \cdot \vec{F}$ at a point is the **net rate of outflow per unit volume** there. Positive divergence means a source, negative means a sink.
- $\nabla \times \vec{F}$ measures the **rotation** of the fluid at a point. For a body rotating with angular velocity $\vec{\omega}$, the velocity field is $\vec{v} = \vec{\omega} \times \vec{r}$ and $\nabla \times \vec{v} = 2\vec{\omega}$.

Solenoidal field: $\nabla \cdot \vec{F} = 0$ everywhere (no sources or sinks; for example the magnetic field, or the flow of an incompressible fluid).

Irrotational field: $\nabla \times \vec{F} = \vec{0}$ everywhere. Then $\vec{F} = \nabla\phi$ for some scalar function ϕ called the **scalar potential**, and the field is **conservative** (Part 4).

Que 3.6 **(a) Find a so that $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal. (b) Find a, b, c so that $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational, and find its scalar potential.**

ANSWER

(a) $\nabla \cdot \vec{F} = 1 + 1 + a = 0 \Rightarrow a = -2.$

(b)

$$\nabla \times \vec{F} = \left(\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z)\right)\hat{i} + \left(\frac{\partial}{\partial z}(x + 2y + az) - \frac{\partial}{\partial x}(4x + cy + 2z)\right)\hat{j} + \left(\frac{\partial}{\partial x}(\right.$$

$$= (c + 1)\hat{i} + (a - 4)\hat{j} + (b - 2)\hat{k}.$$

For an irrotational field all components vanish: $a = 4$, $b = 2$, $c = -1$.

Scalar potential. Now $\vec{F} = (x + 2y + 4z)\hat{i} + (2x - 3y - z)\hat{j} + (4x - y + 2z)\hat{k} = \nabla\phi$, so

$$\frac{\partial\phi}{\partial x} = x + 2y + 4z, \quad \frac{\partial\phi}{\partial y} = 2x - 3y - z, \quad \frac{\partial\phi}{\partial z} = 4x - y + 2z.$$

Integrate the first with respect to x: $\phi = \dfrac{x^2}{2} + 2xy + 4xz + f(y, z)$. Comparing ϕ_y with the second: $2x + f_y = 2x - 3y - z$, so $f = -\dfrac{3y^2}{2} - yz + g(z)$. Comparing ϕ_z with the third: $4x - y + g'(z) = 4x - y + 2z$, so $g = z^2$.

$$\phi = \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy - yz + 4xz + C.$$

Que 3.7 **Prove the following identities for a scalar field ϕ and vector field $\vec{F}$ with continuous second derivatives: (a) $\nabla \times (\nabla \phi) = \vec{0}$, (b) $\nabla \cdot (\nabla \times \vec{F}) = 0$, (c) $\nabla \cdot (\phi \vec{F}) = \nabla \phi \cdot \vec{F} + \phi \nabla \cdot \vec{F}$.**

ANSWER

(a) The $\hat{i}$ component of $\nabla \times \nabla \phi$ is $\dfrac{\partial}{\partial y}\dfrac{\partial \phi}{\partial z} - \dfrac{\partial}{\partial z}\dfrac{\partial \phi}{\partial y} = \phi_{zy} - \phi_{yz} = 0$, by equality of mixed derivatives. The other components vanish in the same way. So **the curl of a gradient is zero**: every gradient field is irrotational.

(b)

$$\nabla \cdot (\nabla \times \vec{F}) = \frac{\partial}{\partial x}\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right).$$

The six terms cancel in pairs ($F_{3,yx} - F_{3,xy}$, and so on), giving 0. So **the divergence of a curl is zero**: every curl field is solenoidal.

(c)

$$\nabla \cdot (\phi \vec{F}) = \sum \frac{\partial}{\partial x}(\phi F_1) = \sum \left(\frac{\partial \phi}{\partial x} F_1 + \phi \frac{\partial F_1}{\partial x}\right) = \nabla \phi \cdot \vec{F} + \phi \nabla \cdot \vec{F}. \blacksquare$$

Que 3.8 **For $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that (a) $\nabla \cdot \vec{r} = 3$ and $\nabla \times \vec{r} = \vec{0}$; (b) $\nabla \cdot \left(\dfrac{\vec{r}}{r^3}\right) = 0$; (c) $\nabla^2 r^n = n(n+1)r^{n-2}$.**

ANSWER

(a) $\nabla \cdot \vec{r} = 1 + 1 + 1 = 3$. The $\hat{i}$ component of $\nabla \times \vec{r}$ is $\dfrac{\partial z}{\partial y} - \dfrac{\partial y}{\partial z} = 0$, and similarly for the others, so $\nabla \times \vec{r} = \vec{0}$.

(b) Using identity (c) of Que 3.7 with $\phi = r^{-3}$ and Que 3.4:

$$\nabla \cdot (r^{-3}\vec{r}) = \nabla(r^{-3}) \cdot \vec{r} + r^{-3}\,\nabla \cdot \vec{r} = (-3r^{-5}\vec{r}) \cdot \vec{r} + 3r^{-3} = -3r^{-5}r^2 + 3r^{-3} = 0.$$

So the inverse-square field $\vec{r}/r^3$ is solenoidal (away from the origin).

(c) $\nabla^2 r^n = \nabla \cdot (\nabla r^n) = \nabla \cdot (n r^{n-2}\vec{r})$. By identity (c):

$$= n\left[\nabla(r^{n-2}) \cdot \vec{r} + r^{n-2} \cdot 3\right] = n\left[(n-2)r^{n-4}\vec{r} \cdot \vec{r} + 3r^{n-2}\right] = n\left[(n-2)r^{n-2} + 3r^{n-2}\right] = n(n+1)r^{n-2}.$$

In particular, $\nabla^2\left(\dfrac{1}{r}\right) = (-1)(0)r^{-3} = 0$: the function $1/r$ satisfies Laplace's equation.

Part 3: Line Integrals and Work Done

Que 3.9 **Define the line integral of a vector field. Find the work done by the force $\vec{F} = x^2\hat{i} + xy\,\hat{j}$ in moving a particle from $(0,0)$ to $(1,1)$ (a) along the parabola $y = x^2$, (b) along the line $y = x$.**

ANSWER

Let C be a curve from A to B given by $\vec{r}(t)$, $a \le t \le b$. The **line integral** of $\vec{F}$ along C is

$$\int_C \vec{F}\cdot d\vec{r} = \int_C (F_1\,dx + F_2\,dy + F_3\,dz) = \int_a^b \vec{F}(\vec{r}(t))\cdot\frac{d\vec{r}}{dt}\,dt.$$

If $\vec{F}$ is a force, this integral is the **work done** in moving a particle along C. If C is closed it is written $\oint_C$ and called the **circulation** of $\vec{F}$ around C.

Method: write x, y, z (and dx, dy, dz) in terms of one parameter using the equation of the curve, and integrate.

(a) Along $y = x^2$: $dy = 2x\,dx$, x from 0 to 1.

$$W = \int_0^1 \left(x^2\,dx + x\cdot x^2\cdot 2x\,dx\right) = \int_0^1 (x^2 + 2x^4)\,dx = \frac{1}{3} + \frac{2}{5} = \frac{11}{15}.$$

(b) Along $y = x$: $dy = dx$.

$$W = \int_0^1 (x^2 + x^2)\,dx = \frac{2}{3}.$$

The work depends on the path, so this force is **not conservative**. Indeed $\nabla\times\vec{F} = \left(\frac{\partial(xy)}{\partial x} - \frac{\partial(x^2)}{\partial y}\right)\hat{k} = y\,\hat{k} \neq \vec{0}$.

Que 3.10 **What is a conservative field? Show that $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is conservative, find its scalar potential, and find the work done in moving a particle from $(1,-2,1)$ to $(3,1,4)$.**

ANSWER

A vector field $\vec{F}$ is **conservative** if the line integral $\int_A^B \vec{F}\cdot d\vec{r}$ depends only on the end points A and B, not on the path. The following are equivalent (in a simply connected region):

1. $\vec{F}$ is conservative.
2. $\oint_C \vec{F}\cdot d\vec{r} = 0$ around every closed curve.
3. $\nabla\times\vec{F} = \vec{0}$ (irrotational).
4. $\vec{F} = \nabla\phi$ for some scalar potential ϕ, and then $\int_A^B \vec{F}\cdot d\vec{r} = \phi(B) - \phi(A)$.

Check.

$$\nabla \times \vec{F} = (0-0)\,\hat{i} + \left(3z^2 - 3z^2\right)\hat{j} + (2x - 2x)\,\hat{k} = \vec{0}.$$

So $\vec{F}$ is conservative.

Potential. $\phi_x = 2xy + z^3 \Rightarrow \phi = x^2y + xz^3 + f(y,z)$. Then $\phi_y = x^2 + f_y = x^2 \Rightarrow f_y = 0$, and $\phi_z = 3xz^2 + f_z = 3xz^2 \Rightarrow f_z = 0$. Hence

$$\phi = x^2y + xz^3 + C.$$

Work done.

$$W = \phi(3,1,4) - \phi(1,-2,1) = (9 + 192) - (-2 + 1) = 201 + 1 = 202.$$

Part 4: Surface and Volume Integrals; Green's Theorem

Que 3.11 **Define surface and volume integrals of vector fields.**

ANSWER

Surface integral (flux). Let S be a surface with unit normal $\hat{n}$ at each point. The **flux** of $\vec{F}$ through S is

$$\iint_S \vec{F} \cdot \hat{n}\, dS.$$

If $\vec{F}$ is the velocity of a fluid, this is the volume of fluid crossing S per unit time. For a closed surface, $\hat{n}$ is taken as the **outward** normal.

Evaluation by projection. If S projects one-to-one onto a region R of the xy-plane, then $dS = \dfrac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$ and

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \iint_R \frac{\vec{F}\cdot\hat{n}}{|\hat{n}\cdot\hat{k}|}\,dx\,dy.$$

(Projection on the yz- or zx-plane works in the same way with $\hat{i}$ or $\hat{j}$.)

Volume integral. For a region V of space, $\iiint_V \phi\, dV$ for a scalar and $\iiint_V \vec{F}\, dV$ for a vector (integrating each component), with $dV = dx\,dy\,dz$.

Que 3.12 **State Green's theorem in the plane and verify it for $\oint_C \left[(x^2+y)\,dx + 3x\,dy\right]$, where C is the boundary of the triangle with vertices $(0,0)$, $(1,0)$ and $(1,1)$.**

ANSWER

Green's theorem. If R is a closed region of the xy-plane bounded by a simple closed curve C traversed **anticlockwise**, and $M(x,y)$, $N(x,y)$ have continuous first partial derivatives on R, then

NEXT ENGINEER _sheerat_

$$\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy.$$

Green's theorem is the plane form of Stokes' theorem. A useful consequence: the area of R is $\frac{1}{2}\oint_C (x\,dy - y\,dx)$.

Verification. $M = x^2 + y$, $N = 3x$.

Right side. $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 3 - 1 = 2$, so the double integral is $2\times$ (area of the triangle) $= 2 \times \frac{1}{2} = 1$.

Left side. Go round anticlockwise: $O(0,0) \to A(1,0) \to B(1,1) \to O$.

- OA: $y = 0$, $dy = 0$, x from 0 to 1: $\int_0^1 x^2\,dx = \frac{1}{3}$.
- AB: $x = 1$, $dx = 0$, y from 0 to 1: $\int_0^1 3\,dy = 3$.
- BO: $y = x$, $dy = dx$, x from 1 to 0: $\int_1^0 \left(x^2 + x + 3x\right) dx = -\left(\frac{1}{3} + 2\right) = -\frac{7}{3}$.

$$\oint_C = \frac{1}{3} + 3 - \frac{7}{3} = 1.$$

Both sides equal 1, so Green's theorem is verified.

Part 5: Stokes' Theorem

Que 3.13 **State Stokes' theorem. Verify it for $\vec{F} = y\,\hat{i} + z\,\hat{j} + x\,\hat{k}$ over the upper half of the sphere $x^2 + y^2 + z^2 = 1$.**

ANSWER

Stokes' theorem. If S is an open, two-sided surface bounded by a simple closed curve C, and $\vec{F}$ has continuous first partial derivatives, then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n}\,dS,$$

where C is traversed in the direction that appears anticlockwise when viewed from the side to which $\hat{n}$ points (right-hand rule).

In words: **the circulation of $\vec{F}$ around a closed curve equals the flux of its curl through any surface spanning that curve.**

Verification. S is the upper hemisphere with outward normal $\hat{n} = x\hat{i} + y\hat{j} + z\hat{k}$ (the radius is 1). Its boundary C is the unit circle $x^2 + y^2 = 1$ in the plane $z = 0$, traversed anticlockwise seen from above.

Line integral. On C: $x = \cos t$, $y = \sin t$, $z = 0$, $0 \le t \le 2\pi$.

$$\oint_C (y\,dx + z\,dy + x\,dz) = \int_0^{2\pi} \sin t\,(-\sin t)\,dt + 0 + 0 = -\int_0^{2\pi} \sin^2 t\,dt = -\pi.$$

Surface integral.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ y & z & x \end{vmatrix} = (0-1)\hat{i} + (0-1)\hat{j} + (0-1)\hat{k} = -\hat{i} - \hat{j} - \hat{k}.$$

$$(\nabla \times \vec{F}) \cdot \hat{n} = -(x + y + z).$$

Project onto the unit disc R: $x^2 + y^2 \le 1$ in the xy-plane, with $\hat{n} \cdot \hat{k} = z$, so $dS = \dfrac{dx\,dy}{z}$:

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS = \iint_R \frac{-(x+y+z)}{z}\, dx\, dy = -\iint_R \frac{x}{z}\, dx\, dy - \iint_R \frac{y}{z}\, dx\, dy - \iint_R dx\, dy.$$

Here $z = \sqrt{1 - x^2 - y^2}$ is even in x and in y, so $\dfrac{x}{z}$ is odd in x and $\dfrac{y}{z}$ is odd in y; over the symmetric disc these two integrals are 0. The last integral is the area of the disc, π.

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS = -\pi.$$

Both sides equal $-\pi$, so Stokes' theorem is verified.

Que 3.14 **Using Stokes' theorem, show that $\oint_C \vec{r} \cdot d\vec{r} = 0$ for any closed curve, and that $\oint_C \phi \nabla\phi \cdot d\vec{r} = 0$.**

ANSWER

(a) $\oint_C \vec{r} \cdot d\vec{r} = \iint_S (\nabla \times \vec{r}) \cdot \hat{n}\, dS = 0$, since $\nabla \times \vec{r} = \vec{0}$ (Que 3.8).

(b) $\phi \nabla\phi = \nabla\left(\dfrac{\phi^2}{2}\right)$ is a gradient, so its curl is zero (Que 3.7(a)). By Stokes' theorem the circulation is 0.

Part 6: Gauss's Divergence Theorem

Que 3.15 **State Gauss's divergence theorem. Verify it for $\vec{F} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ over the cube $0 \le x, y, z \le 1$.**

ANSWER

Gauss's divergence theorem. If V is a closed region of space bounded by a closed surface S with outward unit normal $\hat{n}$, and $\vec{F}$ has continuous first partial derivatives, then

$$\iint_S \vec{F} \cdot \hat{n}\, dS = \iiint_V \nabla \cdot \vec{F}\, dV.$$

In words: **the total outward flux through a closed surface equals the volume integral of the divergence** (the total strength of the sources inside).

Verification.

Volume integral. $\nabla \cdot \vec{F} = 2x + 2y + 2z$.

$$\iiint_V (2x + 2y + 2z)\, dV = 2\int_0^1\int_0^1\int_0^1 (x + y + z)\, dx\, dy\, dz = 2\left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right) = 3.$$

Surface integral. The cube has six faces:

Face	$\hat{n}$	$\vec{F} \cdot \hat{n}$ on the face	Flux
$x = 1$	$\hat{i}$	$x^2 = 1$	$\iint 1\, dy\, dz = 1$
$x = 0$	$-\hat{i}$	$-x^2 = 0$	0
$y = 1$	$\hat{j}$	$y^2 = 1$	1
$y = 0$	$-\hat{j}$	0	0
$z = 1$	$\hat{k}$	$z^2 = 1$	1
$z = 0$	$-\hat{k}$	0	0

Total flux $= 1 + 1 + 1 = 3$.

Both sides equal 3, so the theorem is verified.

Que 3.16 **Using the divergence theorem, evaluate $\iint_S (x\, dy\, dz + y\, dz\, dx + z\, dx\, dy)$ over the sphere $x^2 + y^2 + z^2 = a^2$.**

ANSWER

The integrand is $\vec{F} \cdot \hat{n}\, dS$ with $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k} = \vec{r}$ (because $\hat{n}\, dS$ has components $dy\, dz$, $dz\, dx$, $dx\, dy$). By Gauss's theorem,

$$\iint_S \vec{r} \cdot \hat{n}\, dS = \iiint_V \nabla \cdot \vec{r}\, dV = 3\iiint_V dV = 3 \cdot \frac{4}{3}\pi a^3 = 4\pi a^3.$$

(Check directly: on the sphere $\hat{n} = \vec{r}/a$, so $\vec{r} \cdot \hat{n} = a$, and the surface integral is $a \times 4\pi a^2 = 4\pi a^3$. ✓)

Que 3.17 **Compare the three integral theorems.**

ANSWER

Theorem	Relates	Formula	Used for
Green's (plane)	Line integral around a closed plane curve and double integral over the enclosed region	$\oint_C M\, dx + N\, dy = \iint_R (N_x - M_y)\, dA$	Areas, plane circulation
Stokes'	Line integral around a closed curve and surface integral over any surface it bounds	$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS$	Circulation, conservative fields
Gauss's	Surface integral over a closed surface and volume integral inside it	$\oiint_S \vec{F} \cdot \hat{n}\, dS = \iiint_V \nabla \cdot \vec{F}\, dV$	Flux, continuity equation

All three convert an integral over a boundary into an integral over the region it encloses. Green's theorem is the special case of Stokes' theorem for a flat surface in the xy-plane.

NEXT ENGINEER
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Short-Answer Questions (2 marks)

Q1. Find $\nabla\phi$ for $\phi = x^2 + y^2 + z^2$ at $(1, 1, 1)$.

$2x\hat{i} + 2y\hat{j} + 2z\hat{k} = 2\hat{i} + 2\hat{j} + 2\hat{k}$.

Q2. What is the geometric meaning of $\nabla\phi$?

It is normal to the level surface $\phi = c$ and points in the direction of the greatest rate of increase of ϕ.

Q3. Define a solenoidal vector field.

A field with zero divergence everywhere: $\nabla \cdot \vec{F} = 0$.

Q4. Define an irrotational vector field.

A field with zero curl everywhere: $\nabla \times \vec{F} = \vec{0}$; it can be written as $\nabla\phi$.

Q5. Find the divergence of $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$. $y + z + x$.

Q6. Prove that $\operatorname{curl}(\operatorname{grad}\phi) = \vec{0}$. Each component is a difference of equal mixed partial derivatives, such as $\phi_{zy} - \phi_{yz} = 0$.

Q7. Find $\nabla \cdot \vec{r}$ and $\nabla \times \vec{r}$. 3 and $\vec{0}$.

Q8. Write the condition for a force field to be conservative.

$\nabla \times \vec{F} = \vec{0}$ (equivalently $\vec{F} = \nabla\phi$, or zero work around every closed path).

Q9. State Gauss's divergence theorem.

$\oiint_S \vec{F} \cdot \hat{n}\, dS = \iiint_V \nabla \cdot \vec{F}\, dV$ for a closed surface S enclosing the volume V.

Q10. State Stokes' theorem.

$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS$, where S is any surface bounded by C.

Q11. Write the area of a region using Green's theorem.

$A = \dfrac{1}{2}\oint_C (x\, dy - y\, dx)$.

Q12. Find the maximum directional derivative of $\phi = xyz$ at $(1, 1, 1)$.

$\nabla\phi = \hat{i} + \hat{j} + \hat{k}$, so the maximum is $\sqrt{3}$.

Practice Questions

1. Find the directional derivative of $\phi = x^2 - y^2 + 2z^2$ at $P(1, 2, 3)$ in the direction of PQ where $Q = (5, 0, 4)$. (Answer: $\frac{28}{\sqrt{21}}$.)
2. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$. (Answer: $\cos^{-1}\frac{8}{3\sqrt{21}}$.)
3. Show that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational.
4. Prove that $\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \nabla^2\vec{F}$.
5. Find the work done by $\vec{F} = 3xy\hat{i} - y^2\hat{j}$ along the curve $y = 2x^2$ from $(0, 0)$ to $(1, 2)$. (Answer: $-\frac{7}{6}$.)

NEXT ENGINEER _sheerat_

6. Show that $\vec{F} = (y+z)\hat{i} + (z+x)\hat{j} + (x+y)\hat{k}$ is conservative and find its potential. (Answer: $\phi = xy + yz + zx$.)
7. Verify Green's theorem for $\oint (xy\,dx + x^2\,dy)$ around the boundary of the square $0 \le x, y \le 1$. (Both sides: $\frac{1}{2}$.)
8. Using Green's theorem, find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (Answer: πab.)
9. Verify Stokes' theorem for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\,\hat{j}$ over the rectangle $0 \le x \le a$, $0 \le y \le b$ in the xy-plane. (Both sides: $-2ab^2$.)
10. Using the divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n}\, dS$ for $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ over the region bounded by $x^2 + y^2 = 4$, $z = 0$ and $z = 3$. (Answer: 84π.)

NEXT ENGINEER
sheerat

UNIT 4

Matrices

Unit overview

Matrices are the language of linear algebra. They store systems of linear equations, represent transformations of space, and are at the heart of computer graphics, machine learning, circuit analysis and structural engineering. This unit covers special complex matrices, elementary transformations, rank, the solution of linear systems (directly by Gauss elimination and iteratively by Gauss–Seidel), eigenvalues and eigenvectors, and diagonalization.

Topics in this unit: symmetric, skew-symmetric, Hermitian, skew-Hermitian, orthogonal and unitary matrices; elementary row transformations; inverse by Gauss–Jordan; rank by echelon form; consistency of linear systems; Gauss elimination; Gauss–Seidel iteration; eigenvalues and eigenvectors and their properties; Cayley–Hamilton theorem; similar matrices and diagonalization.

Most asked: rank by echelon form, consistency of a system with parameters, Gauss–Seidel iterations, eigenvalues and eigenvectors of a 3 × 3 matrix, Cayley–Hamilton theorem with inverse, and diagonalization.

Part 1: Special Types of Matrices

Que 4.1 **Define symmetric, skew-symmetric, Hermitian, skew-Hermitian, orthogonal and unitary matrices with examples.**

ANSWER

For a matrix A, A^T is its transpose, $\bar{A}$ the matrix of complex conjugates, and $A^\theta = (\bar{A})^T$ the **conjugate transpose** (also written A^*).

Matrix	Condition	Diagonal elements	Example
Symmetric	$A^T = A$	any real	$\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix}$
Skew-symmetric	$A^T = -A$	all zero	$\begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}$
Hermitian	$A^\theta = A$	all real	$\begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$
Skew-Hermitian	$A^\theta = -A$	zero or purely imaginary	$\begin{bmatrix} 2i & 1+i \\ -1+i & 0 \end{bmatrix}$
Orthogonal	$AA^T = A^TA = I$	—	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$
Unitary	$AA^\theta = A^\theta A = I$	—	$\frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$

A real Hermitian matrix is symmetric, and a real unitary matrix is orthogonal.

Verification of the unitary example. Let $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$. Then $A^\theta = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} = A$, and

$$AA^\theta = \frac{1}{3}\begin{bmatrix} 1+(1+i)(1-i) & (1+i)-(1+i) \\ (1-i)-(1-i) & (1-i)(1+i)+1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = I.$$

Important results

1. Every square matrix is uniquely the sum of a symmetric and a skew-symmetric matrix: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.
2. Every square matrix is uniquely the sum of a Hermitian and a skew-Hermitian matrix: $A = \frac{1}{2}(A + A^\theta) + \frac{1}{2}(A - A^\theta)$.
3. The eigenvalues of a Hermitian (or real symmetric) matrix are **real**; of a skew-Hermitian matrix are **zero or purely imaginary**; of a unitary (or orthogonal) matrix have **modulus 1**.
4. The inverse and the transpose of an orthogonal matrix are orthogonal; the product of two orthogonal matrices is orthogonal.

Que 4.2 Express $A = \begin{bmatrix} 2+3i & 1-i \\ 3 & 4i \end{bmatrix}$ as the sum of a Hermitian and a skew-Hermitian matrix.

ANSWER

$$A^\theta = \overline{A}^T = \begin{bmatrix} 2-3i & 3 \\ 1+i & -4i \end{bmatrix}.$$

Hermitian part.

$$P = \frac{1}{2}(A + A^\theta) = \frac{1}{2}\begin{bmatrix} 4 & 4-i \\ 4+i & 0 \end{bmatrix} = \begin{bmatrix} 2 & \frac{4-i}{2} \\ \frac{4+i}{2} & 0 \end{bmatrix}.$$

Skew-Hermitian part.

$$Q = \frac{1}{2}(A - A^\theta) = \frac{1}{2}\begin{bmatrix} 6i & -2-i \\ 2-i & 8i \end{bmatrix} = \begin{bmatrix} 3i & \frac{-2-i}{2} \\ \frac{2-i}{2} & 4i \end{bmatrix}.$$

Check. $P^\theta = P$ (real diagonal, conjugate off-diagonal entries), and $Q^\theta = \begin{bmatrix} -3i & \frac{2+i}{2} \\ \frac{-2+i}{2} & -4i \end{bmatrix} = -Q$. Also $P + Q = A$.

Que 4.3 Prove that the eigenvalues of a Hermitian matrix are real.

ANSWER

Let $AX = \lambda X$ with $X \neq 0$, where $A^\theta = A$. Multiply on the left by X^θ:

$$X^\theta AX = \lambda X^\theta X. \tag{1}$$

NEXT ENGINEER _sheerat_

Take the conjugate transpose of (1). The left side becomes $X^\theta A^\theta X = X^\theta A X$, and the right side becomes $\bar{\lambda} X^\theta X$:

$$X^\theta A X = \bar{\lambda} X^\theta X. \tag{2}$$

From (1) and (2), $(\lambda - \bar{\lambda}) X^\theta X = 0$. Since $X^\theta X = \sum |x_i|^2 > 0$, we get $\lambda = \bar{\lambda}$, so λ is real. ∎

Example: the Hermitian matrix $\begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$ has characteristic equation $\lambda^2 - 5\lambda + (6-2) = 0$, giving the real eigenvalues $\lambda = 1, 4$.

Part 2: Elementary Transformations, Inverse and Rank

Que 4.4 **What are elementary row transformations? Find the inverse of $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ by the Gauss–Jordan method.**

ANSWER

Elementary row transformations are:

1. Interchange of two rows: $R_i \leftrightarrow R_j$.
2. Multiplication of a row by a non-zero constant: $R_i \to kR_i$.
3. Adding a multiple of one row to another: $R_i \to R_i + kR_j$.

They do not change the rank of a matrix or the solution set of the corresponding system. Two matrices obtained from each other by elementary transformations are called **equivalent** ($A \sim B$).

Gauss–Jordan method for A^{-1}: write the augmented matrix $[A \mid I]$ and apply row transformations until the left part becomes I; the right part is then A^{-1}.

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 3 & 2 & 3 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \to R_2 - 3R_1$, $R_3 \to R_3 - R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -4 & 0 & -3 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right]$$

$R_2 \to -\frac{1}{4} R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{4} & -\frac{1}{4} & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right]$$

$R_1 \to R_1 - 2R_2$, $R_3 \to R_3 + R_2$:

NEXT ENGINEER _sheerat_

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & \frac{3}{4} & -\frac{1}{4} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & 1 \end{array}\right]$$

$R_1 \to R_1 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{4} & \frac{3}{4} & -1 \\ 0 & 1 & 0 & \frac{3}{4} & -\frac{1}{4} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & 1 \end{array}\right]$$

$$A^{-1} = \frac{1}{4}\begin{bmatrix} -1 & 3 & -4 \\ 3 & -1 & 0 \\ -1 & -1 & 4 \end{bmatrix}.$$

Check. The first row of A times the first column of A^{-1}: $\frac{1}{4}(-1+6-1) = 1$. The other products give the identity similarly. Also $|A| = -4 \neq 0$, so A is invertible.

Que 4.5 **Define the rank of a matrix and the echelon form. Find the rank of** $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$.

ANSWER

Rank. The rank $\rho(A)$ of a matrix is the order of its largest non-zero minor (the largest square sub-matrix with non-zero determinant). Equivalently, it is the maximum number of linearly independent rows (or columns).

Echelon form. A matrix is in (row) echelon form if

1. all zero rows are at the bottom,
2. the first non-zero entry (the **pivot**) of each row is to the right of the pivot of the row above, and
3. all entries below a pivot are zero.

The **rank equals the number of non-zero rows** in an echelon form.

Solution.

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$, $R_4 \to R_4 - 6R_1$:

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -8 & 3 \\ 0 & -4 & -11 & 5 \end{bmatrix}$$

$R_2 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -11 & 5 \end{bmatrix}$$

NEXT ENGINEER _sheerat_

$R_4 \to R_4 - R_2$:

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & -3 & 2 \end{bmatrix}$$

$R_4 \to R_4 - R_3$:

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are three non-zero rows, so $\rho(A) = 3$. (Indeed $R_4 = R_1 + R_2 + R_3$ in the original matrix.)

Normal form. By using column transformations as well, every matrix of rank r can be reduced to the **normal form** $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$. This gives another way to find the rank.

Part 3: Systems of Linear Equations

Que 4.6 **Explain the conditions for consistency of a system of linear equations $AX = B$.**

ANSWER

A system of m equations in n unknowns is written $AX = B$, where A is the $m \times n$ **coefficient matrix**, and $[A \mid B]$ is the **augmented matrix**. The system is **consistent** if it has at least one solution.

Rouché–Capelli theorem

Condition	Result
$\rho(A) \neq \rho([A \mid B])$	inconsistent: no solution
$\rho(A) = \rho([A \mid B]) = n$ (number of unknowns)	unique solution
$\rho(A) = \rho([A \mid B]) = r < n$	infinitely many solutions, with $n - r$ free variables

Homogeneous system $AX = 0$: it is always consistent ($X = 0$ is a solution, the **trivial solution**). It has **non-trivial solutions** if and only if $\rho(A) < n$. For a square matrix, this means $|A| = 0$.

Que 4.7 **Investigate for what values of λ and μ the system $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ has (i) no solution, (ii) a unique solution, (iii) infinitely many solutions.**

ANSWER

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]$$

$R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{array}\right]$$

$R_3 \to R_3 - R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{array}\right]$$

(i) No solution: if $\lambda = 3$ and $\mu \neq 10$, the last row reads $0 = \mu - 10 \neq 0$; then $\rho(A) = 2 \neq \rho([A \mid B]) = 3$.

(ii) Unique solution: if $\lambda \neq 3$ (μ any value), $\rho(A) = \rho([A \mid B]) = 3 = n$.

(iii) Infinitely many solutions: if $\lambda = 3$ and $\mu = 10$, $\rho(A) = \rho([A \mid B]) = 2 < 3$. Taking $z = k$: $y = 4 - 2k$, $x = 6 - y - z = 2 + k$.

Que 4.8 **Solve by the Gauss elimination method:** $x + y + z = 6$, $2x + 3y + 4z = 20$, $3x + 2y + 5z = 22$.

ANSWER

Gauss elimination reduces the augmented matrix to an upper triangular (echelon) form using row operations, and then finds the unknowns by **back substitution**.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 4 & 20 \\ 3 & 2 & 5 & 22 \end{array}\right]$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -1 & 2 & 4 \end{array}\right]$$

$R_3 \to R_3 + R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 4 & 12 \end{array}\right]$$

Back substitution: $4z = 12 \Rightarrow z = 3$; $y + 2z = 8 \Rightarrow y = 2$; $x + y + z = 6 \Rightarrow x = 1$.

Solution: $x = 1$, $y = 2$, $z = 3$. (Check in the third equation: $3 + 4 + 15 = 22$. ✓)

Que 4.9 **Find the non-trivial solutions of** $x + y + z = 0$, $2x - y + 3z = 0$, $3x + 4z = 0$.

ANSWER

NEXT ENGINEER _sheerat_

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 0 & 4 \end{vmatrix} = 1(-4-0) - 1(8-9) + 1(0+3) = -4+1+3 = 0.$$

Since $|A| = 0$, non-trivial solutions exist. Reduce:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 0 & 4 \end{bmatrix} \xrightarrow{R_2 - 2R_1,\ R_3 - 3R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & 1 \\ 0 & -3 & 1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$\rho(A) = 2 < 3$, so there is $3 - 2 = 1$ free variable. From $-3y + z = 0$, $z = 3y$. Put $y = k$: $z = 3k$, and $x = -y - z = -4k$.

$$x = -4k, \quad y = k, \quad z = 3k \quad (k \text{ arbitrary}).$$

Que 4.10 **Explain the Gauss–Seidel iteration method. Solve $10x - 2y + z = 19$, $x + 8y - 2z = 8$, $2x - y + 10z = 13$.**

ANSWER

Gauss–Seidel method. It is an **iterative** method: start with a guess and improve it repeatedly.

1. Arrange the equations so that the system is **diagonally dominant**: in each equation, the absolute value of the diagonal coefficient is at least the sum of the absolute values of the other coefficients (strictly greater in at least one). This guarantees convergence.
2. Solve the first equation for x, the second for y, the third for z.
3. Start with $x = y = z = 0$ (or any guess). Compute the new x, then **immediately use this new** x in computing y, and the new x, y in computing z. (Using the latest values is what distinguishes Gauss–Seidel from the Jacobi method and makes it converge faster.)
4. Repeat until two successive iterations agree to the required accuracy.

Solution. The system is diagonally dominant ($10 > 2+1$, $8 > 1+2$, $10 > 2+1$).

$$x = \frac{1}{10}(19 + 2y - z), \qquad y = \frac{1}{8}(8 - x + 2z), \qquad z = \frac{1}{10}(13 - 2x + y).$$

Iteration	x	y	z
0	0	0	0
1	1.9000	0.7625	0.9963
2	1.9529	1.0050	1.0099
3	2.0000	1.0025	1.0002
4	2.0005	1.0000	0.9999
5	2.0000	1.0000	1.0000

NEXT ENGINEER

Working of iteration 1: $x = \frac{1}{10}(19 + 0 - 0) = 1.9$; $y = \frac{1}{8}(8 - 1.9 + 0) = 0.7625$; $z = \frac{1}{10}(13 - 3.8 + 0.7625) = 0.99625$.

Solution: $x = 2$, $y = 1$, $z = 1$ (correct to four decimal places). Check: $20 - 2 + 1 = 19$. ✓

Part 4: Eigenvalues and Eigenvectors

Que 4.11 **Define eigenvalues and eigenvectors. Find the eigenvalues and eigenvectors of** $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$.

ANSWER

Definition. For a square matrix A of order n, a scalar λ is an **eigenvalue** (characteristic root) if there is a non-zero vector X with

$$AX = \lambda X.$$

X is an **eigenvector** belonging to λ. Geometrically, A only stretches X by the factor λ; it does not change its direction.

$AX = \lambda X$ means $(A - \lambda I)X = 0$, which has a non-zero solution only if

$$|A - \lambda I| = 0.$$

This is the **characteristic equation**. For a 3×3 matrix it is

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0,$$

where S_1 = trace (sum of diagonal elements) and S_2 = sum of the minors of the diagonal elements.

Solution. $S_1 = 2 + 2 + 2 = 6$. $S_2 = (4 - 0) + (4 - 1) + (4 - 0) = 11$. $|A| = 2(4) - 0 + 1(0 - 2) = 6$.

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0 \;\Rightarrow\; (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0.$$

Eigenvalues: $\lambda = 1, 2, 3$.

$\lambda = 1$**:** $(A - I)X = 0$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \;\Rightarrow\; x + z = 0,\; y = 0 \;\Rightarrow\; X_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

$\lambda = 2$**:** $(A - 2I)X = 0$:

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \;\Rightarrow\; z = 0,\; x = 0,\; y \text{ free} \;\Rightarrow\; X_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

NEXT ENGINEER _sheerat_

$\lambda = 3$: $(A - 3I)X = 0$:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \Rightarrow x = z,\ y = 0 \Rightarrow X_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

(Any non-zero multiple of an eigenvector is also an eigenvector.) Since A is symmetric, the three eigenvectors are mutually orthogonal: $X_1 \cdot X_2 = 0$, $X_1 \cdot X_3 = -1 + 1 = 0$, $X_2 \cdot X_3 = 0$.

Que 4.12 **State the important properties of eigenvalues.**

ANSWER

Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of an $n \times n$ matrix A.

1. $\lambda_1 + \lambda_2 + \cdots + \lambda_n = \text{trace}(A)$.
2. $\lambda_1 \lambda_2 \cdots \lambda_n = |A|$. Hence A is singular if and only if 0 is an eigenvalue.
3. A and A^T have the same eigenvalues.
4. The eigenvalues of A^{-1} are $\dfrac{1}{\lambda_i}$; of kA are $k\lambda_i$; of A^m are λ_i^m; of $A + kI$ are $\lambda_i + k$. The eigenvectors stay the same.
5. The eigenvalues of a triangular or diagonal matrix are its diagonal elements.
6. Eigenvectors belonging to **distinct** eigenvalues are linearly independent.
7. For a real symmetric matrix, eigenvalues are real and eigenvectors of distinct eigenvalues are orthogonal.
8. **Algebraic multiplicity** of λ = the number of times it is a root of the characteristic equation; **geometric multiplicity** = the number of independent eigenvectors for it. Geometric multiplicity ≤ algebraic multiplicity.

Example. For the matrix in Que 4.11, $1 + 2 + 3 = 6$ = trace and $1 \cdot 2 \cdot 3 = 6 = |A|$. The eigenvalues of A^{-1} are $1, \frac{1}{2}, \frac{1}{3}$ and of A^2 are $1, 4, 9$.

Que 4.13 **State the Cayley–Hamilton theorem. Verify it for** $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ **and hence find** A^{-1}.

ANSWER

Cayley–Hamilton theorem: every square matrix satisfies its own characteristic equation. If $|A - \lambda I| = \lambda^n + a_1\lambda^{n-1} + \cdots + a_n = 0$, then

$$A^n + a_1 A^{n-1} + \cdots + a_n I = 0.$$

Characteristic equation. $S_1 = 6$. $S_2 = (4 - 1) + (4 - 1) + (4 - 1) = 9$. $|A| = 2(4 - 1) + 1(-2 + 1) + 1(1 - 2) = 6 - 1 - 1 = 4$.

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0.$$

Verification.

$$A^2 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}, \qquad A^3 = A^2A = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}.$$

(For example, the $(1,1)$ entry of A^3 is $6 \cdot 2 + (-5)(-1) + 5 \cdot 1 = 22$.)

$$A^3 - 6A^2 + 9A - 4I = \begin{bmatrix} 22 - 36 + 18 - 4 & -21 + 30 - 9 & 21 - 30 + 9 \\ \cdots & \cdots & \cdots \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

All nine entries reduce to zero in the same way, so the theorem is verified.

Inverse. Multiply $A^3 - 6A^2 + 9A - 4I = 0$ by A^{-1}:

$$A^2 - 6A + 9I - 4A^{-1} = 0 \;\Rightarrow\; A^{-1} = \frac{1}{4}\left(A^2 - 6A + 9I\right).$$

$$A^2 - 6A + 9I = \begin{bmatrix} 6 - 12 + 9 & -5 + 6 & 5 - 6 \\ -5 + 6 & 6 - 12 + 9 & -5 + 6 \\ 5 - 6 & -5 + 6 & 6 - 12 + 9 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

$$A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

Part 5: Similarity and Diagonalization

Que 4.14 **Define similar matrices and diagonalization. State the condition for a matrix to be diagonalizable.**

ANSWER

Similar matrices. B is similar to A if $B = P^{-1}AP$ for some non-singular matrix P. Similar matrices have the same characteristic equation, hence the same eigenvalues, trace, determinant and rank. (Proof: $|P^{-1}AP - \lambda I| = |P^{-1}(A - \lambda I)P| = |P^{-1}||A - \lambda I||P| = |A - \lambda I|$.)

Diagonalization. A is **diagonalizable** if it is similar to a diagonal matrix D:

$$P^{-1}AP = D = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

Condition. An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors. In particular, a matrix with n **distinct eigenvalues** is always diagonalizable, and every **real symmetric** matrix is diagonalizable.

Method.

1. Find the eigenvalues $\lambda_1, \ldots, \lambda_n$ and eigenvectors $X_1, \ldots, X_n$.
2. Form the **modal matrix** $P = [X_1\ X_2\ \cdots\ X_n]$ (eigenvectors as columns).
3. Then $P^{-1}AP = D$, with the eigenvalues on the diagonal in the same order as the columns.

For a real symmetric matrix, normalising the eigenvectors makes P **orthogonal**, so $P^{-1} = P^T$ and $P^TAP = D$ (orthogonal diagonalization).

Use: powers of A become easy: $A^m = PD^mP^{-1}$.

Que 4.15 **Diagonalize $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ and hence find A^n.**

ANSWER

Eigenvalues. $|A - \lambda I| = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5) = 0$, so $\lambda = 2, 5$.

Eigenvectors.

- $\lambda = 2$: $\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} X = 0 \Rightarrow 2x + y = 0 \Rightarrow X_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$.
- $\lambda = 5$: $\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} X = 0 \Rightarrow x = y \Rightarrow X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Modal matrix.

$$P = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}, \qquad |P| = -3, \qquad P^{-1} = \frac{1}{-3}\begin{bmatrix} 1 & -1 \\ -2 & -1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}.$$

Check. $AP = \begin{bmatrix} -2 & 5 \\ 4 & 5 \end{bmatrix} = PD$ with $D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$, so $P^{-1}AP = D$.

Powers.

$$A^n = PD^nP^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2^n & 0 \\ 0 & 5^n \end{bmatrix} \cdot \frac{1}{3}\begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2^n + 2\cdot 5^n & -2^n + 5^n \\ -2\cdot 2^n + 2\cdot 5^n & 2\cdot 2^n + 5^n \end{bmatrix}.$$

Check with $n = 1$: $\frac{1}{3}\begin{bmatrix} 2+10 & -2+5 \\ -4+10 & 4+5 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = A$. ✓

Que 4.16 **Orthogonally diagonalize the symmetric matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$.**

ANSWER

From Que 4.11 the eigenvalues are $1, 2, 3$ with orthogonal eigenvectors $(-1, 0, 1)$, $(0, 1, 0)$, $(1, 0, 1)$. Normalise them (divide by their lengths $\sqrt{2}$, 1, $\sqrt{2}$):

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

P is orthogonal ($P^TP = I$), so $P^{-1} = P^T$ and

$$P^T AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Orthogonal diagonalization is used to reduce a **quadratic form** $X^T AX = 2x^2 + 2y^2 + 2z^2 + 2xz$ to the **canonical form** $u^2 + 2v^2 + 3w^2$ by the substitution $X = PY$.

NEXT ENGINEER
sheerat

Short-Answer Questions (2 marks)

Q1. Define a Hermitian matrix with an example.

$A^\theta = A$. Example: $\begin{bmatrix} 1 & 2+i \\ 2-i & 4 \end{bmatrix}$.

Q2. Show that the diagonal elements of a skew-Hermitian matrix are zero or purely imaginary.

$a_{ii} = -\bar{a}_{ii}$ gives $a_{ii} + \bar{a}_{ii} = 2\,\mathrm{Re}(a_{ii}) = 0$.

Q3. Define the rank of a matrix.

The order of the largest non-vanishing minor; equivalently, the number of non-zero rows in its echelon form.

Q4. When does $AX = B$ have a unique solution?

When $\rho(A) = \rho([A \mid B]) = n$, the number of unknowns.

Q5. When does a homogeneous system have a non-trivial solution?

When $\rho(A) < n$; for a square system, when $|A| = 0$.

Q6. What is the condition for convergence of the Gauss–Seidel method?

The coefficient matrix should be diagonally dominant.

Q7. If the eigenvalues of A are $2, 3, 4$, find the eigenvalues of A^{-1} and A^2.

$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ and $4, 9, 16$.

Q8. Find the eigenvalues of $\begin{bmatrix} 3 & 5 & 7 \\ 0 & -2 & 1 \\ 0 & 0 & 4 \end{bmatrix}$. It is upper triangular, so the eigenvalues are $3, -2, 4$.

Q9. Two eigenvalues of a 3×3 matrix with trace 9 and determinant 24 are 2 and 3. Find the third.

From the trace: $9 - 5 = 4$ (and $2 \cdot 3 \cdot 4 = 24$ ✓).

Q10. State the Cayley–Hamilton theorem.

Every square matrix satisfies its own characteristic equation.

Q11. Define similar matrices.

$B = P^{-1}AP$ for some non-singular P; they have the same eigenvalues.

Q12. Is every square matrix diagonalizable?

No. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has the repeated eigenvalue 1 with only one independent eigenvector, so it is not diagonalizable.

Practice Questions

1. Show that $A = \frac{1}{2}\begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$ is unitary.

NEXT ENGINEER _sheerat_

2. Find the rank of $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$. (Answer: 3.)
3. Find the inverse of $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$ by elementary transformations. (Answer: $\begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}$.)
4. Test the consistency and solve: $2x - y + 3z = 8$, $-x + 2y + z = 4$, $3x + y - 4z = 0$. (Answer: $x = y = z = 2$.)
5. For what value of k does $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$ have a solution? Solve in each case. (Answer: $k = 1$ or $k = 2$.)
6. Solve by Gauss–Seidel: $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$. (Answer: $1, -1, 1$.)
7. Find the eigenvalues and eigenvectors of $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$. (Eigenvalues: $-2, 3, 6$.)
8. Verify the Cayley–Hamilton theorem for $\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and find A^4. (Answer: $A^2 = 5I$, so $A^4 = 25I$.)
9. Diagonalize $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$. (Eigenvalues: $1, 2, 3$.)
10. Prove that the eigenvalues of a unitary matrix have absolute value 1.

NEXT ENGINEER
sheerat

UNIT 5

Introduction to Vector Spaces

Unit overview

In school a vector is an arrow with magnitude and direction. Linear algebra keeps only the two operations that make arrows useful, **adding** them and **multiplying by numbers**, and studies every collection of objects that can be added and scaled in the same way. Polynomials, matrices, solutions of differential equations and the data vectors of machine learning all turn out to be vectors in this sense. The ideas of this unit (subspace, linear independence, basis, dimension) are the foundation of everything in linear algebra.

Topics in this unit: fields, vector spaces and examples, subspaces, linear combinations and span, linear dependence and independence, basis and dimension, coordinates, sum and direct sum of subspaces.

Most asked: prove that a given set is (or is not) a vector space or subspace, test vectors for linear dependence, express a vector as a linear combination, find a basis and the dimension of a subspace, and the dimension formula for the sum of subspaces.

Part 1: Fields and Vector Spaces

Que 5.1 **Define a field. Give examples and non-examples.**

ANSWER

A **field** F is a set with two operations, addition $(+)$ and multiplication $(\cdot)$, such that for all $a, b, c \in F$:

Addition

1. Closure: $a + b \in F$.
2. Associative: $(a + b) + c = a + (b + c)$.
3. Commutative: $a + b = b + a$.
4. There is a zero element 0 with $a + 0 = a$.
5. Every a has an additive inverse $-a$ with $a + (-a) = 0$.

Multiplication

6. Closure: $ab \in F$.
7. Associative: $(ab)c = a(bc)$.
8. Commutative: $ab = ba$.
9. There is a unit element $1 \neq 0$ with $a \cdot 1 = a$.
10. Every $a \neq 0$ has a multiplicative inverse a^{-1} with $aa^{-1} = 1$.

Distributive law

11. $a(b+c) = ab + ac$.

Examples: the rational numbers $\mathbb{Q}$, the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$, and $\mathbb{Z}_p = \{0, 1, \ldots, p-1\}$ with arithmetic modulo a prime p.

Non-examples: the integers $\mathbb{Z}$ (2 has no multiplicative inverse in $\mathbb{Z}$); the natural numbers $\mathbb{N}$ (no additive inverses); $\mathbb{Z}_6$ (since $2 \cdot 3 = 0$, the element 2 has no inverse).

The elements of the field are called **scalars**. In this course the field is $\mathbb{R}$ (real vector spaces) or $\mathbb{C}$ (complex vector spaces).

Que 5.2 **Define a vector space. Give standard examples.**

ANSWER

A **vector space** over a field F is a non-empty set V with two operations:

- **vector addition:** $u + v \in V$ for $u, v \in V$, and
- **scalar multiplication:** $\alpha u \in V$ for $\alpha \in F$, $u \in V$,

satisfying the following axioms for all $u, v, w \in V$ and $\alpha, \beta \in F$:

No.	Axiom	Name
1	$u + v \in V$	closure under addition
2	$u + v = v + u$	commutativity
3	$(u + v) + w = u + (v + w)$	associativity
4	there is $0 \in V$ with $u + 0 = u$	zero vector
5	for each u there is $-u$ with $u + (-u) = 0$	additive inverse
6	$\alpha u \in V$	closure under scalar multiplication
7	$\alpha(u + v) = \alpha u + \alpha v$	distributive (over vectors)
8	$(\alpha + \beta)u = \alpha u + \beta u$	distributive (over scalars)
9	$(\alpha\beta)u = \alpha(\beta u)$	associativity of scaling
10	$1 \cdot u = u$	unit property

Standard examples

1. $\mathbb{R}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{R}\}$ with componentwise addition and scaling.
2. P_n, the set of all real polynomials of degree at most n, with the usual addition and multiplication by numbers.
3. $M_{m \times n}$, the set of all $m \times n$ real matrices, with matrix addition and scalar multiplication.
4. $C[a, b]$, the set of all continuous real functions on $[a, b]$, with $(f + g)(x) = f(x) + g(x)$ and $(\alpha f)(x) = \alpha f(x)$.
5. The set of solutions of a homogeneous linear differential equation such as $y'' + y = 0$.

Basic consequences of the axioms: for every $u \in V$ and scalar α,

$$0 \cdot u = 0, \qquad \alpha \cdot 0 = 0, \qquad (-1)u = -u, \qquad \alpha u = 0 \Rightarrow \alpha = 0 \text{ or } u = 0.$$

Proof of the first: $0 \cdot u = (0+0)u = 0 \cdot u + 0 \cdot u$; adding $-(0 \cdot u)$ to both sides gives $0 = 0 \cdot u$.

Que 5.3 **Show that the set V of all 2×2 real matrices of the form $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ is a vector space over $\mathbb{R}$, and that the set of all polynomials of degree exactly 2 is not.**

ANSWER

(a) V is a subset of $M_{2\times 2}$, which is a vector space. So it is enough to check closure (the remaining axioms are inherited; see the subspace test in Que 5.4).

- Zero matrix: $a = b = 0$ gives $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in V$.
- Addition: $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} + \begin{bmatrix} c & d \\ -d & c \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ -(b+d) & a+c \end{bmatrix} \in V$.
- Scaling: $\alpha \begin{bmatrix} a & b \\ -b & a \end{bmatrix} = \begin{bmatrix} \alpha a & \alpha b \\ -\alpha b & \alpha a \end{bmatrix} \in V$.

Hence V is a vector space. (These matrices behave exactly like complex numbers $a + ib$.)

(b) Let W be the set of polynomials of degree exactly 2. Then $p(x) = x^2 + x$ and $q(x) = -x^2 + 1$ are in W, but

$$p(x) + q(x) = x + 1,$$

which has degree 1, so it is not in W. Closure under addition fails, and W also does not contain the zero polynomial. So W is **not** a vector space.

Part 2: Subspaces

Que 5.4 **Define a subspace. State and prove the necessary and sufficient condition for a non-empty subset to be a subspace.**

ANSWER

A non-empty subset W of a vector space V is a **subspace** of V if W is itself a vector space under the same operations.

Theorem (subspace test). A non-empty subset W of V is a subspace if and only if

$$\alpha u + \beta v \in W \quad \text{for all } u, v \in W \text{ and all scalars } \alpha, \beta.$$

(Equivalently: W is closed under addition and under scalar multiplication.)

Proof.

Necessity: if W is a subspace it is closed under both operations, so $\alpha u, \beta v \in W$ and therefore $\alpha u + \beta v \in W$.

Sufficiency: suppose the condition holds.

- Taking $\alpha = \beta = 1$: $u + v \in W$ (closure under addition).
- Taking $\beta = 0$: $\alpha u \in W$ (closure under scaling).
- Taking $\alpha = \beta = 0$: $0 \in W$ (zero vector).
- Taking $\alpha = -1$, $\beta = 0$: $-u \in W$ (inverses).
- The remaining axioms (commutativity, associativity, distributive laws, $1 \cdot u = u$) hold for all vectors of V, so they hold in particular for vectors of W.

Hence W is a vector space, that is, a subspace. ∎

Quick check: every subspace must contain the zero vector. If $0 \notin W$, then W is not a subspace.

Every vector space V has two trivial subspaces: $\{0\}$ and V itself.

Que 5.5 **Which of the following are subspaces of $\mathbb{R}^3$? (a) $W_1 = \{(x, y, z) : x + y + z = 0\}$, (b) $W_2 = \{(x, y, z) : x + y + z = 1\}$, (c) $W_3 = \{(x, y, z) : x \geq 0\}$, (d) $W_4 = \{(x, y, z) : x = 2y\}$.**

ANSWER

(a) Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ be in W_1, so $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$. Then $\alpha u + \beta v = (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2)$, and the sum of its components is

$$\alpha(x_1 + y_1 + z_1) + \beta(x_2 + y_2 + z_2) = 0 + 0 = 0.$$

So $\alpha u + \beta v \in W_1$: W_1 **is a subspace** (a plane through the origin).

(b) $(0, 0, 0) \notin W_2$ since $0 + 0 + 0 \neq 1$. **Not a subspace** (a plane not passing through the origin).

(c) $u = (1, 0, 0) \in W_3$, but $(-1)u = (-1, 0, 0) \notin W_3$. Not closed under scalar multiplication, so **not a subspace**.

(d) If $x_1 = 2y_1$ and $x_2 = 2y_2$, then $\alpha x_1 + \beta x_2 = 2(\alpha y_1 + \beta y_2)$. W_4 **is a subspace.**

General rule: the solution set of a **homogeneous** system of linear equations $AX = 0$ is always a subspace (the **null space** of A); the solution set of a non-homogeneous system $AX = B$, $B \neq 0$, never is.

Que 5.6 **Prove that the intersection of two subspaces is a subspace, but their union need not be.**

ANSWER

Intersection. Let W_1, W_2 be subspaces of V. $0 \in W_1$ and $0 \in W_2$, so $0 \in W_1 \cap W_2$; the intersection is non-empty. If $u, v \in W_1 \cap W_2$, then $\alpha u + \beta v \in W_1$ (as W_1 is a subspace) and $\alpha u + \beta v \in W_2$. Hence $\alpha u + \beta v \in W_1 \cap W_2$, which is therefore a subspace. ∎

Union. In $\mathbb{R}^2$, let W_1 be the x-axis $\{(x, 0)\}$ and W_2 the y-axis $\{(0, y)\}$. Both are subspaces. But $(1, 0) \in W_1$ and $(0, 1) \in W_2$, while

$$(1, 0) + (0, 1) = (1, 1) \notin W_1 \cup W_2.$$

So $W_1 \cup W_2$ is not closed under addition and is **not a subspace**. (In fact $W_1 \cup W_2$ is a subspace only when one of them contains the other.)

Part 3: Linear Combinations and Span

Que 5.7 **Define linear combination and linear span. Express $v = (4, 3, 5)$ as a linear combination of $u_1 = (1, 1, 0)$, $u_2 = (1, 0, 1)$, $u_3 = (0, 1, 1)$.**

ANSWER

A vector v is a **linear combination** of $u_1, \ldots, u_k$ if there are scalars $c_1, \ldots, c_k$ such that

$$v = c_1 u_1 + c_2 u_2 + \cdots + c_k u_k.$$

The **linear span** of a set $S = \{u_1, \ldots, u_k\}$, written $\text{span}(S)$ or $L(S)$, is the set of **all** linear combinations of vectors of S. It is always a subspace, and it is the **smallest** subspace containing S. If $\text{span}(S) = V$, we say S **spans** (generates) V.

Solution. We need $c_1(1,1,0) + c_2(1,0,1) + c_3(0,1,1) = (4,3,5)$:

$$c_1 + c_2 = 4, \qquad c_1 + c_3 = 3, \qquad c_2 + c_3 = 5.$$

Adding the three equations: $2(c_1 + c_2 + c_3) = 12$, so $c_1 + c_2 + c_3 = 6$. Subtracting each equation from this: $c_3 = 2$, $c_2 = 3$, $c_1 = 1$.

$$(4, 3, 5) = 1 \cdot (1, 1, 0) + 3 \cdot (1, 0, 1) + 2 \cdot (0, 1, 1).$$

Check: $(1 + 3 + 0,\ 1 + 0 + 2,\ 0 + 3 + 2) = (4, 3, 5)$. ✓

Que 5.8 **Is $(1, 2, 4)$ in the span of $(1, 0, 1)$ and $(0, 1, 1)$? Describe the span geometrically.**

ANSWER

$a(1,0,1) + b(0,1,1) = (a, b, a + b)$. For this to equal $(1, 2, 4)$ we need $a = 1$, $b = 2$, and $a + b = 4$. But $1 + 2 = 3 \neq 4$. The system is inconsistent, so $(1, 2, 4)$ **is not in the span**.

The span is $\{(a, b, a + b)\}$, that is, all vectors (x, y, z) with $z = x + y$. Geometrically it is the **plane** $x + y - z = 0$ through the origin. The vector $(1, 2, 3)$ lies in this plane; $(1, 2, 4)$ does not.

Part 4: Linear Dependence and Independence

Que 5.9 **Define linearly dependent and linearly independent sets. State the important results about them.**

ANSWER

Vectors $u_1, u_2, \ldots, u_k$ are **linearly dependent (LD)** if there exist scalars $c_1, \ldots, c_k$, **not all zero**, such that

$$c_1u_1 + c_2u_2 + \cdots + c_ku_k = 0.$$

They are **linearly independent (LI)** if this equation holds **only** when $c_1 = c_2 = \cdots = c_k = 0$.

Important results

1. A set is LD if and only if at least one of its vectors is a linear combination of the others.
2. Any set containing the zero vector is LD (take coefficient 1 on the zero vector and 0 on the rest).
3. Two vectors are LD if and only if one is a scalar multiple of the other.
4. Any subset of an LI set is LI; any superset of an LD set is LD.
5. In $\mathbb{R}^n$, any set of more than n vectors is LD.
6. **Determinant/rank test:** n vectors of $\mathbb{R}^n$ are LI if and only if the determinant of the matrix having them as rows (or columns) is non-zero. More generally, k vectors are LI if and only if the matrix formed by them has rank k.
7. Eigenvectors belonging to distinct eigenvalues are LI (Unit 4).

Que 5.10 **Test for linear dependence: (a) $(1,2,1)$, $(2,1,0)$, $(1,-1,2)$; (b) $(1,2,3)$, $(2,3,4)$, $(3,4,5)$. In case of dependence, find the relation.**

ANSWER

(a)

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & -1 & 2 \end{vmatrix} = 1(2-0) - 2(4-0) + 1(-2-1) = 2 - 8 - 3 = -9 \neq 0.$$

The vectors are **linearly independent**.

(b)

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 1(15-16) - 2(10-12) + 3(8-9) = -1 + 4 - 3 = 0.$$

The vectors are **linearly dependent**. To find the relation, solve $c_1(1,2,3) + c_2(2,3,4) + c_3(3,4,5) = 0$:

$$c_1 + 2c_2 + 3c_3 = 0, \qquad 2c_1 + 3c_2 + 4c_3 = 0.$$

(The third equation follows from these two.) Subtracting twice the first from the second: $-c_2 - 2c_3 = 0$, so $c_2 = -2c_3$, and then $c_1 = -2c_2 - 3c_3 = 4c_3 - 3c_3 = c_3$. Taking $c_3 = 1$: $c_1 = 1$, $c_2 = -2$.

$$(1,2,3) - 2(2,3,4) + (3,4,5) = 0, \quad \text{that is} \quad (3,4,5) = 2(2,3,4) - (1,2,3).$$

NEXT ENGINEER _sheerat_

Que 5.11 **Show that the functions 1, x, x^2 are linearly independent in P_2, and that $\{1+x,\ x+x^2,\ 1+2x+x^2\}$ is linearly dependent.**

ANSWER

(a) If $c_1 \cdot 1 + c_2 x + c_3 x^2 = 0$ for all x (the zero polynomial), then all coefficients of the polynomial must be zero: $c_1 = c_2 = c_3 = 0$. So $\{1, x, x^2\}$ is LI.

(b) Notice that $(1+x) + (x+x^2) = 1 + 2x + x^2$. So

$$1 \cdot (1+x) + 1 \cdot (x+x^2) - 1 \cdot (1+2x+x^2) = 0,$$

with coefficients not all zero. The set is **LD**.

(Alternatively, write each polynomial by its coefficient vector: $(1,1,0)$, $(0,1,1)$, $(1,2,1)$. The determinant is $1(1-2) - 1(0-1) + 0 = 0$, confirming dependence.)

Part 5: Basis and Dimension

Que 5.12 **Define basis and dimension. Give the standard bases of $\mathbb{R}^n$, P_n and $M_{2\times 2}$.**

ANSWER

A subset $B = \{v_1, \ldots, v_n\}$ of a vector space V is a **basis** of V if

1. B is **linearly independent**, and
2. B **spans** V.

Every vector of V can then be written **uniquely** as $v = c_1 v_1 + \cdots + c_n v_n$. The numbers $(c_1, \ldots, c_n)$ are the **coordinates** of v relative to B.

Dimension. All bases of a vector space have the same number of elements. This number is the **dimension** of V, written $\dim V$. If $V = \{0\}$, $\dim V = 0$. A space with a finite basis is **finite-dimensional**.

Standard bases

Space	Standard basis	Dimension
$\mathbb{R}^n$	$e_1 = (1,0,\ldots,0),\ \ldots,\ e_n = (0,\ldots,0,1)$	n
P_n	$1, x, x^2, \ldots, x^n$	$n+1$
$M_{2\times 2}$	$\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&1\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}$	4
$M_{m\times n}$	the mn matrices with a single 1	mn

Key theorems (for $\dim V = n$):

1. Any n linearly independent vectors form a basis.
2. Any n vectors that span V form a basis.
3. Any LI set can be **extended** to a basis; any spanning set can be **reduced** to a basis.

4. Every subspace W of V satisfies $\dim W \leq n$, with equality only when $W = V$.

Que 5.13 **Show that $B = \{(1,1,0),(1,0,1),(0,1,1)\}$ is a basis of $\mathbb{R}^3$ and find the coordinates of $(4,3,5)$ relative to B.**

ANSWER

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 1(0-1) - 1(1-0) + 0 = -2 \neq 0.$$

So the three vectors are LI. Since $\dim \mathbb{R}^3 = 3$, three LI vectors form a basis (Que 5.12, theorem 1).

From Que 5.7, $(4,3,5) = 1(1,1,0) + 3(1,0,1) + 2(0,1,1)$, so the **coordinate vector** of $(4,3,5)$ relative to B is $(1,3,2)$.

Que 5.14 **Find a basis and the dimension of the subspace $W = \{(x,y,z) \in \mathbb{R}^3 : x + y + z = 0\}$.**

ANSWER

From $x + y + z = 0$, $x = -y - z$, with y and z free. So every vector of W is

$$(x,y,z) = (-y-z,\ y,\ z) = y(-1,1,0) + z(-1,0,1).$$

Thus $W = \text{span}\{(-1,1,0),(-1,0,1)\}$. These two vectors are LI (neither is a multiple of the other). Hence

$$\text{Basis of } W = \{(-1,1,0),\ (-1,0,1)\}, \qquad \dim W = 2.$$

General method: for the solution space of a homogeneous system $AX = 0$ in n unknowns, $\dim(\text{solution space}) = n - \rho(A)$. Here $n = 3$, $\rho(A) = 1$, so the dimension is 2. ✓

Que 5.15 **Find a basis and the dimension of the subspace of $\mathbb{R}^3$ spanned by $(1,2,3)$, $(2,3,4)$, $(3,4,5)$.**

ANSWER

Write the vectors as rows and reduce to echelon form (row operations do not change the span of the rows):

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{R_2 - 2R_1,\ R_3 - 3R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -2 & -4 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The non-zero rows form a basis: $\{(1,2,3),\ (0,1,2)\}$ (the second row multiplied by -1). **Dimension** $= 2$, which agrees with Que 5.10(b): the three vectors are dependent, and the span is a plane through the origin.

Que 5.16 **Find a basis and the dimension of (a) the space of all 2×2 symmetric matrices, (b) $W = \{p(x) \in P_2 : p(1) = 0\}$.**

ANSWER

(a) A symmetric 2×2 matrix has the form

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

The three matrices on the right are LI and span the space, so they form a basis, and the **dimension is 3**. (In general, $n \times n$ symmetric matrices form a space of dimension $\frac{n(n+1)}{2}$.)

(b) Let $p(x) = a + bx + cx^2$. The condition $p(1) = 0$ gives $a + b + c = 0$, so $a = -b - c$ and

$$p(x) = -b - c + bx + cx^2 = b(x - 1) + c(x^2 - 1).$$

So $W = \text{span}\{x - 1,\ x^2 - 1\}$, and these two polynomials are LI (different degrees). **Basis** $\{x - 1, x^2 - 1\}$**, dimension 2.**

Part 6: Sum and Direct Sum of Subspaces

Que 5.17 **Define the sum and direct sum of subspaces. State the dimension theorem and illustrate it.**

ANSWER

Sum. If W_1 and W_2 are subspaces of V, their sum is

$$W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1,\ w_2 \in W_2\}.$$

It is a subspace, and it is the smallest subspace containing both W_1 and W_2.

Direct sum. V is the **direct sum** $W_1 \oplus W_2$ if $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$. Then every $v \in V$ can be written **uniquely** as $v = w_1 + w_2$.

Dimension theorem. For finite-dimensional subspaces,

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

For a direct sum, $\dim(W_1 \oplus W_2) = \dim W_1 + \dim W_2$.

Example 1. In $\mathbb{R}^3$, let $W_1 = \{(x, y, 0)\}$ (the xy-plane) and $W_2 = \{(0, y, z)\}$ (the yz-plane). Every $(a, b, c) = (a, b, 0) + (0, 0, c)$, so $W_1 + W_2 = \mathbb{R}^3$. Their intersection is the y-axis $\{(0, y, 0)\}$, of dimension 1. Check:

$$\dim(W_1 + W_2) = 2 + 2 - 1 = 3. \checkmark$$

The sum is **not direct**, because the intersection is not $\{0\}$; for example $(1,1,1) = (1,1,0) + (0,0,1) = (1,0,0) + (0,1,1)$ in two different ways.

Example 2. Let W_1 be the xy-plane and $W_3 = \{(0,0,z)\}$ (the z-axis). Then $W_1 \cap W_3 = \{0\}$ and $W_1 + W_3 = \mathbb{R}^3$, so

$$\mathbb{R}^3 = W_1 \oplus W_3, \qquad 3 = 2 + 1.$$

Example 3. Every square matrix is uniquely the sum of a symmetric and a skew-symmetric matrix (Unit 4), so $M_{n\times n} = (\text{symmetric}) \oplus (\text{skew-symmetric})$. For $n = 2$: $4 = 3 + 1$.

NEXT ENGINEER
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Short-Answer Questions (2 marks)

Q1. Define a vector space.

A non-empty set V with vector addition and scalar multiplication over a field F, closed under both operations and satisfying the ten axioms (commutativity, associativity, zero, inverses, distributive laws, $1 \cdot u = u$).

Q2. Is $\mathbb{Z}$ a field? Why?

No. Non-zero integers such as 2 have no multiplicative inverse in $\mathbb{Z}$.

Q3. Is $\{(x, y) : y = 2x + 1\}$ a subspace of $\mathbb{R}^2$?

No, it does not contain $(0, 0)$.

Q4. State the subspace test.

A non-empty $W \subseteq V$ is a subspace iff $\alpha u + \beta v \in W$ for all $u, v \in W$ and all scalars α, β.

Q5. When is a set of vectors linearly dependent?

When some non-trivial linear combination of them equals the zero vector.

Q6. Are $(1, 2)$ and $(2, 4)$ linearly independent?

No: $(2, 4) = 2(1, 2)$.

Q7. Define basis.

A linearly independent set that spans the vector space.

Q8. What is the dimension of P_3 and of $M_{2\times 3}$?

4 and 6.

Q9. Can four vectors in $\mathbb{R}^3$ be linearly independent?

No. Any set of more than 3 vectors in $\mathbb{R}^3$ is linearly dependent.

Q10. What is the dimension of the solution space of $x - y + 2z = 0$ in $\mathbb{R}^3$?

$3 - 1 = 2$.

Q11. Define the direct sum of subspaces.

$V = W_1 \oplus W_2$ if $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$.

Q12. Write the dimension formula for the sum of two subspaces.

$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$.

Practice Questions

1. Show that the set of all real-valued functions f on $[0, 1]$ with $f(0) = 0$ is a vector space, but the set with $f(0) = 1$ is not.
2. Show that $W = \{A \in M_{2\times 2} : \text{trace}(A) = 0\}$ is a subspace and find its dimension. (Answer: 3.)

NEXT ENGINEER _sheerat_

3. Is $\{(x, y, z) : x^2 + y^2 = z^2\}$ a subspace of $\mathbb{R}^3$? Justify. (Answer: no; $(1,0,1) + (0,1,1) = (1,1,2)$ is not in it.)
4. Express $(2, -5, 3)$ as a linear combination of $(1, -3, 2)$, $(2, -4, -1)$, $(1, -5, 7)$ if possible. (Answer: not possible.)
5. Test for dependence: $(1,1,1,1)$, $(0,1,1,1)$, $(0,0,1,1)$, $(0,0,0,1)$. (Answer: independent.)
6. Find k so that $(1,2,1)$, $(k,1,1)$, $(0,1,1)$ are dependent. (Answer: $k = 0$.)
7. Extend $\{(1,1,1)\}$ to a basis of $\mathbb{R}^3$. (One answer: add $(0,1,0)$ and $(0,0,1)$.)
8. Find a basis and the dimension of the solution space of $x + 2y - z = 0$, $2x + 4y - 2z = 0$. (Answer: dimension 2; basis $(-2,1,0)$, $(1,0,1)$.)
9. If W_1 and W_2 are two distinct 2-dimensional subspaces of $\mathbb{R}^3$, prove that $\dim(W_1 \cap W_2) = 1$.
10. Show that $\{1,\ 1+x,\ (1+x)^2\}$ is a basis of P_2 and find the coordinates of $3 + 2x + x^2$ relative to it. (Answer: $(2,0,1)$.)

NEXT ENGINEER
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Model Question Paper with Answer Guide

This paper follows the university pattern (70 marks, 3 hours). Next to each question is the place in this book where the full answer is given. Attempt it in 3 hours with the book closed, then check your answers.

Section A: attempt all questions (7 × 2 = 14 marks)

No.	Question	Answer in
a	If $u = x^3 + y^3 + 3x^2y$, find $xu_x + yu_y$.	Que 1.6 (answer $3u$)
b	Find $\dfrac{\partial(r,\theta)}{\partial(x,y)}$ for polar coordinates.	Unit 1, Q5
c	Evaluate $\Gamma\left(\frac{7}{2}\right)$.	Unit 2, Q5
d	Find the divergence of $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$.	Unit 3, Q5
e	Find the directional derivative of $\phi = xyz$ at $(1,1,1)$ along $\hat{i}$.	Que 3.2 method (answer 1)
f	If the eigenvalues of A are $2, 3, 4$, find those of A^{-1}.	Unit 4, Q7
g	Define basis of a vector space.	Unit 5, Q7

Section B: attempt any three (3 × 7 = 21 marks)

No.	Question	Answer in
a	If $u = \sin^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$, prove $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = -\dfrac{\sin u \cos 2u}{4\cos^3 u}$.	Que 1.8
b	Change the order of integration and evaluate $\displaystyle\int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$.	Que 2.4
c	Verify Stokes' theorem for $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ over the upper unit hemisphere.	Que 3.13
d	Verify the Cayley–Hamilton theorem for a 3×3 matrix and find its inverse.	Que 4.13
e	Find a basis and the dimension of $\{(x,y,z) : x + y + z = 0\}$.	Que 5.14

Section C: attempt one part of each question (5 × 7 = 35 marks)

No.	Question	Answer in
3 (a)	Find the extreme values of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.	Que 1.20
3 (b)	Find the dimensions of an open box of volume 32 with least surface area.	Que 1.23
4 (a)	Evaluate $\displaystyle\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx\, dy$ and deduce $\displaystyle\int_0^\infty e^{-x^2} dx$.	Que 2.7
4 (b)	Using Dirichlet's integral, find the volume of an ellipsoid.	Que 2.19
5 (a)	Find a, b, c for an irrotational field and its scalar potential.	Que 3.6
5 (b)	Verify Gauss's divergence theorem over a unit cube.	Que 3.15

No.	Question	Answer in
6 (a)	Discuss the consistency of a system containing parameters λ, μ.	Que 4.7
6 (b)	Solve a system by the Gauss–Seidel method.	Que 4.10
7 (a)	Test vectors for linear dependence and find the relation.	Que 5.10
7 (b)	State and prove the subspace test; show that a union of subspaces need not be a subspace.	Que 5.4, 5.6

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Last-Day Revision Checklist

- Euler's theorem: $xu_x + yu_y = nu$; second order $n(n-1)u$; corollaries $g(u) = n\dfrac{f(u)}{f'(u)}$ and $g(u)[g'(u) - 1]$.
- Total derivative and chain rule; implicit derivative $-\dfrac{f_x}{f_y}$.
- Jacobians: polar r, spherical $r^2 \sin\theta$; $JJ' = 1$; $J = 0$ means functional dependence.
- Maxima and minima table using $rt - s^2$; Lagrange equations $f_x + \lambda\phi_x = 0$.
- Percentage error: powers multiply the percentage errors, which then add.
- Change of order: always sketch the region; split it if one strip formula does not cover it.
- $dx\,dy = r\,dr\,d\theta$; $dV = r^2 \sin\theta\,dr\,d\theta\,d\phi$.
- $\Gamma(n+1) = n\Gamma(n)$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, $B(m,n) = \dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$, the $\sin^p \cos^q$ formula, Dirichlet's integral.
- $\nabla\phi$, $\nabla \cdot \vec{F}$, $\nabla \times \vec{F}$; solenoidal and irrotational; $\nabla r^n = nr^{n-2}\vec{r}$; curl grad $= 0$, div curl $= 0$.
- Green's, Stokes' and Gauss's theorems: statements and one verification each.
- Rank by echelon form; Rouché–Capelli conditions; Gauss elimination and Gauss–Seidel.
- Characteristic equation $\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$; eigenvector method; Cayley–Hamilton inverse; diagonalization $P^{-1}AP = D$, $A^n = PD^nP^{-1}$.
- Subspace test; LD/LI by determinant; basis and dimension; $\dim(W_1 + W_2)$ formula.

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